Article

Terrain-Aware AUV Survey-Line Planning for Multibeam Bathymetric Mapping Using Public Bathymetry Benchmarks

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Abstract

AUV-assisted bathymetric surveying is a core capability for autonomous maritime systems operating in complex seafloor environments, but the pre-mission line plans used for multibeam echo sounder (MBES) mapping are often built from uniform-spacing assumptions that ignore terrain-dependent swath variation. This paper develops a terrain-aware pre-mission survey-line planning framework for public bathymetry-based mapping. The proposed planner combines a local sensor–terrain MBES footprint model, a discrete orientation scan, quantile-based adaptive spacing, and Genetic Algorithm (GA) refinement of cross-track line positions while preserving an inspectable lawnmower sweep order. Evaluation uses two public GEBCO 2025 gridded bathymetry subsets as the primary benchmark, three synthetic terrains as controlled stress tests, a higher-resolution USGS Southern Cascadia 30 m multibeam-derived public-grid extension, a coarse-prior/fine-grid public replay, and an execution-uncertainty replay with cross-track, heading, and footprint-scale perturbations. Results on the two GEBCO public scenes show that the hybrid planner attains 98.97–99.63 percent predicted coverage and yields a modest mean path-length shortening of 0.75 percent, while its primary scientific value is the much stronger suppression of mean scene-level excess-overlap violation from about 0.81 percent to 0.095 percent. The ablation shows that most public route shortening comes from terrain-aware spacing, whereas GA refinement mainly suppresses residual overlap and stabilizes the final line layout across 20 seeds. The USGS high-complexity crop and the coarse-prior replay both reproduce the high-overlap regime in which geometry-aware spacing is most beneficial, and the uncertainty replay shows that the public layouts remain mostly feasible under moderate perturbations but require additional execution margin under stronger noise. The study provides an auditable public-bathymetry benchmark and a predictable prior-map geometry layer for subsequent survey-grade evaluation and execution-aware AUV mission planning.

Keywords: AUV-assisted bathymetric surveying; Autonomous maritime systems; Multibeam echo sounder (MBES); Pre-mission survey-line planning; Public bathymetry benchmark; Coverage path planning; Genetic Algorithm (GA)

1. Introduction

Accurate seafloor mapping supports ocean engineering, marine resource management, habitat assessment, and navigation safety in complex maritime operations. Multibeam echosounders (MBES) are central to modern bathymetric survey because each pass observes a wide cross-track strip of seafloor, enabling dense terrain reconstruction at far higher efficiency than single-beam sampling [3,4]. When an MBES is carried by an Autonomous Underwater Vehicle (AUV), the mission plan becomes part of the sensing system: survey-line orientation, placement, and spacing determine how the acoustic footprint samples the seabed.

AUV-assisted bathymetric surveying is especially attractive for autonomous maritime systems because the platform can follow repeatable line patterns over shelf breaks, canyon margins, and other terrain where surface-vessel access, sea state, or safety constraints may limit conventional operations. The usual lawnmower pattern is operationally attractive because it is easy to inspect, approve, and execute, but uniform spacing is poorly matched to irregular seafloor. Effective MBES swath width varies with depth, slope, and beam incidence, so a spacing that is adequate in one part of a scene can be too conservative elsewhere or insufficient where the footprint contracts.

Coverage path planning (CPP) for bathymetric survey is therefore a geometry-aware pre-mission planning problem. A planner must choose the orientation, placement, and spacing of survey lines so that the union of MBES footprints covers the target region while limiting total path length and redundant overlap [6,7,22]. Recent bathymetric survey planners already span dedicated coastal and offshore bathymetric mapping formulations, energy-constrained coverage, and learning-based underwater path planning [10,11,32–34]. Within that broader design space, a focused line-layout question remains important for repeatable AUV mission preparation: how should a fixed parallel-track family be oriented and spaced when the prior bathymetry already reveals strong terrain-dependent swath variation?

Adjacent underwater CPP studies have also advanced online replanning, cooperative multi-vehicle coverage, and heuristic optimization under vehicle or environmental constraints [26–31,36–41]. These directions are important for autonomous maritime operations, but a geometry-agnostic baseline line family can still carry avoidable sensing mismatch into later route sequencing, online correction, or fleet coordination. This paper therefore addresses the pre-mission geometry layer: given a prior bathymetric map and a known MBES configuration, can terrain-dependent sensing geometry improve an inspectable fixed-pattern AUV survey plan?

The proposed planner embeds the sensor–terrain model directly into the line-layout search. A greedy stage scans candidate survey orientations, and a second stage refines the cross-track line positions with a Genetic Algorithm (GA) while preserving the fixed sweep order. Candidate layouts are evaluated by predicted coverage, excess overlap above the prescribed overlap ceiling, total path length, and feasibility. This design keeps the traversal pattern deliberately simple while making the spacing decision responsive to the bathymetry that controls the effective MBES footprint.

The main contributions of this paper are as follows:

- We formulate a terrain-aware pre-mission CPP problem for AUV-assisted bathymetric surveying under a fixed lawnmower traversal, explicitly modeling how effective MBES swath width varies with local seafloor geometry instead of assuming constant line spacing.
- We develop a geometry-aware planner in which a discrete orientation scan and quantile-based adaptive spacing determine the baseline line family, and a Genetic Algorithm refines the cross-track line positions while preserving an operator-inspectable fixed sweep pattern.
- We introduce a reproducible public bathymetry-based benchmark that combines two GEBCO 2025 subset scenes with three synthetic stress-test terrains, allowing geometry-aware and fixed-spacing layouts to be compared with the same sensor–terrain evaluator.

- We show on the public-bathymetry benchmark that most route-length savings come from terrain-aware spacing itself, whereas the full hybrid GA mainly improves overlap cleanup and final-layout repeatability; relative to Fixed-Spacing, it reduces mean scene-level excess-overlap violation from about 0.81 percent to 0.095 percent and achieves a 0.75 percent mean scene-level path-length shortening.
- We add sensitivity, extension, coarse-prior replay, and uncertainty-replay evidence, including target-overlap, prior-map perturbation, grid-resolution diagnostics, a separate USGS Southern Cascadia 30 m multibeam-derived public-grid check, and Monte Carlo execution perturbations, to connect public-data benchmark behavior with the requirements for survey-grade transfer.
- We expose a boundary-of-validity result that is useful for mission planning: the hardest synthetic complex-relief scene remains infeasible under the present single-heading formulation, while 20-seed runs on the two public scenes demonstrate strong seed-level repeatability for the stochastic refinement stage.

The intended application is civilian pre-mission bathymetric survey planning for ocean engineering and marine mapping, where a transparent fixed-line plan is often easier to review than a fully adaptive route. The study addresses AUV survey-line geometry with an available prior terrain model and evaluates performance on public-scene benchmarks, synthetic stress tests, a higher-resolution public multibeam-derived extension, and a Monte Carlo uncertainty replay. Online replanning, full vehicle dynamics, current-driven control, communication constraints, and route-order optimization beyond the fixed sweep pattern are complementary execution-aware modules that can build on the fixed-line geometry layer evaluated here.

Accordingly, the evaluation is organized around six linked questions. First, does terrain-aware spacing improve a fixed parallel-track survey on real public gridded bathymetry? Second, is the public-scene gain caused mainly by adaptive spacing or by the stochastic refinement stage? Third, where does the single-heading lawnmower assumption fail? Fourth, does the same pattern remain visible on an independent higher-resolution public bathymetry grid? Fifth, what happens when a line family is planned from a coarser public prior and then replayed on a finer grid without re-optimization? Sixth, how do the selected public layouts behave when cross-track, heading, and footprint-scale uncertainty is injected after planning? These questions motivate the separation between the public GEBCO results, the public-scene ablation, the synthetic complex-terrain failure case, the USGS extension, the coarse-prior/fine-grid replay, and the uncertainty-replay check.

The remainder of the paper first positions this fixed-pattern problem against adjacent AUV coverage, mapping, and navigation literature, then details the sensor–terrain model and line-layout optimizer, and finally reports public-grid, synthetic, sensitivity, extension, coarse-prior replay, and uncertainty-replay evidence.

2. Related Work and Positioning

Coverage planning for underwater survey has developed along several related but distinct lines. The first line studies online or intermittent replanning for AUV bathymetric mapping. In this setting, the vehicle may update its path as new measurements arrive, and the planner is judged by its ability to respond to incomplete or changing map information [3,7]. This is a natural formulation when the objective is exploration or adaptive mapping. More recent active bathymetric SLAM work goes further by deciding when an AUV should revisit terrain-rich areas to secure valid loop closures during online exploration [8]. Recent underwater mapping papers also widen the adjacent landscape with feature-based bathymetric SLAM, massively parallel Gaussian-process bathymetric modeling, and modular acoustic graph SLAM pipelines for monitoring-oriented deployments [12–15]. The present paper assumes a prior bathymetric surface and asks a complementary pre-mission design question: before the vehicle is deployed, how should a fixed parallel-track family be oriented and spaced if the effective MBES footprint is terrain dependent? This distinction matters because an offline line family remains valuable for operator review, mission approval, and repeatable benchmark comparison.

A second line focuses on track-spacing adaptation and sonar-aware coverage. Yordanova et al. [4] showed that AUV track spacing can be adapted to improve seabed survey data quality, and related ladder-survey work optimized AUV routes while accounting for seabed type and sonar range [5]. Dedicated bathymetric survey planners have since become more explicit about platform and survey geometry, from coastal and offshore bathymetric coverage formulations with practical mapping demonstrations [10,11] to prior-map benthic survey design in feature space under deployment constraints [9]. More recent multi-AUV work has also coupled coverage decisions with sonar-performance models and environmental effects such as terrain and currents [31]. These studies motivate the central modeling premise of this paper: the sensor footprint should not be treated as a constant-radius disk or a constant-width strip when the seafloor geometry changes across the workspace. We embed a terrain-dependent swath model into a fixed-pattern survey-line design problem and measure how much benefit can be obtained without changing the simple lawnmower traversal assumption.

A third line studies cooperative and heuristic optimization for underwater coverage. Recent examples include coverage search of static underwater targets [32], energy-constrained online coverage [34], deep-reinforcement-learning-based AUV path planning [33], multi-AUV search with side-scan sonar [6], cooperative 3-D coverage using swarm migration genetic algorithms [26], ant-colony planning for underwater gliders [28], DDQN-based cooperative navigation-oriented coverage planning [35], and biologically inspired or neural coverage planners for marine vehicles [36,37,39]. These methods broaden the operational design space by adding vehicle cooperation, task allocation, environmental search objectives, or richer path topologies. They also make it harder to isolate the specific effect of terrain-aware MBES line spacing, because changes in route ordering, vehicle count, allocation, and sensor model may occur simultaneously. The benchmark in this paper deliberately keeps the traversal topology fixed so that the comparison isolates three design variables: global survey orientation, cross-track line placement, and nonuniform spacing.

Uncertainty-aware planning is another important direction. Bathymetric survey performance can be affected by SLAM uncertainty, positioning error, prior-map mismatch, vehicle tracking error, sound-speed variation, and attitude disturbance [22,40,41]. Related navigation and mapping studies emphasize terrain-aided localization, prior-map matching, Gaussian-process bathymetric matching, robust particle filtering, and constrained-set navigation under bathymetric priors [16–21,23–25]. These issues motivate the execution-uncertainty replay added in this study: after the public-scene line families are planned, cross-track, heading, and footprint-scale perturbations are injected and the same coverage metrics are recomputed. This remains a numerical stress check, but it links the public benchmark to the kinds of tracking and footprint uncertainty emphasized in the broader literature.

Table 1 summarizes the resulting positioning. The important point is not that prior work lacks sophistication; rather, much of it answers a different question. Online SLAM and terrain-aided navigation papers target localization and map consistency, cooperative CPP papers target allocation or search efficiency, and sonar-performance CPP papers often change the sensing or vehicle model. This paper deliberately keeps the vehicle count, traversal topology, and route order fixed so that the effect of terrain-dependent MBES footprint geometry can be isolated.

Table 1. Positioning of the present fixed-pattern benchmark relative to adjacent AUV coverage, mapping, and navigation literature.

Research line	Representative focus	What it establishes	Gap left for this paper
Online bathymetric mapping and active SLAM	Intermittent replanning and active revisit/exploration decisions [3,7,8,12,14]	The vehicle can adapt online when map or localization information changes.	Does not isolate pre-mission placement of a fixed parallel-line family.
Track-spacing and sonar-aware survey planning	Adaptive spacing and ladder-style survey routes [4,5,10,11]	Track spacing should respond to sonar quality, seabed type, or survey geometry.	Public gridded bathymetry evidence and GA ablation for fixed MBES lawnmower layouts remain limited.
Field-oriented prior-map benthic planning	Feature-space exploration using broad-scale bathymetric priors [9]	Prior bathymetry can guide informative initial survey design under deployment constraints.	Optimizes information-gathering templates rather than MBES coverage overlap and line placement.
Multi-vehicle and sonar-performance CPP	Cooperative allocation, currents, and environment-sensitive sonar range [6,26,31,32,34,35]	Coverage quality can improve when vehicle cooperation, currents, or sonar performance models are included.	Vehicle count, allocation, and route topology change simultaneously, obscuring the fixed-line layout effect.
Terrain-aided navigation and bathymetric SLAM	Localization, prior-map matching, and robust map-consistency management [16–21,23–25]	Bathymetry can support underwater navigation and map correction.	These works are downstream of, or parallel to, offline survey-line geometry design.
This paper	Prior-map fixed-pattern MBES survey-line layout on public GEBCO grids, synthetic stress tests, and a USGS public-grid extension	Terrain-dependent swath geometry is isolated under a fixed lawnmower traversal, with adaptive-spacing, GA-refinement, and post-planning uncertainty-replay effects separated.	Provides an auditable pre-mission line-layout benchmark that can be extended to mission-log replay, field execution, and online autonomy studies.

This positioning defines the role of the contribution. The planner supplies a pre-mission geometry layer that can precede online replanning, cooperative allocation, SLAM-aware survey design, or execution-aware vehicle control. The central question is deliberately measurable: when a prior bathymetric grid is available, does terrain-aware MBES geometry improve the placement of parallel survey lines relative to uniform spacing, and does GA refinement add value beyond adaptive spacing alone? The public GEBCO scenes and synthetic stress tests in Section 4 are chosen to answer that question under a controlled fixed-pattern protocol.

3. Methods

We study offline survey-line design for a single AUV equipped with an MBES and operating with a prior bathymetric model. The execution pattern is fixed to a lawnmower traversal so that the methodological focus is the placement of parallel survey lines under terrain-dependent sensing; route ordering, online correction, and multi-vehicle coordination can be layered onto this fixed-line design in subsequent planners. Figure 1 summarizes the workflow: prior bathymetry and sensor settings define the local swath model, the model informs a global orientation search, and the selected orientation is passed to adaptive spacing and GA refinement before the resulting layout is evaluated.

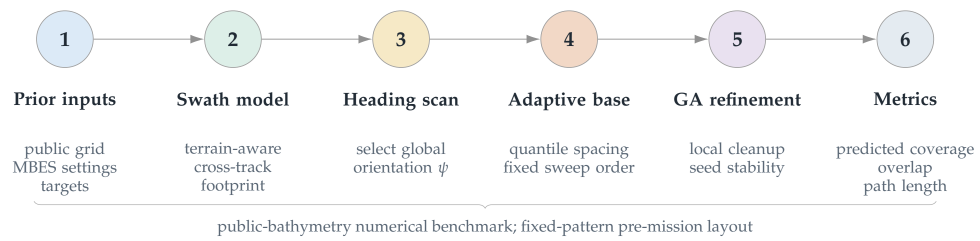

Figure 1. Method workflow for the proposed terrain-aware AUV survey-line planner. A prior bathymetric surface and MBES configuration are converted into terrain-dependent swath predictions; a greedy search selects the global survey orientation ψ; adaptive spacing and GA refinement then place the parallel lines; and the final layout is scored by path length, coverage, excess overlap, feasibility, and seed repeatability.

This section introduces the sensor–terrain geometric model used to estimate local coverage and overlap from a candidate survey line, and then describes the two-stage optimizer built on top of that model. The first stage performs a greedy search over global survey orientation. The second stage applies a GA to refine the positions and spacings of the parallel lines while preserving the fixed sweep structure.

Efficient bathymetric coverage over irregular seabed requires an explicit mapping from each candidate line to the terrain region that can actually be observed. The constant-swath approximation used in many CPP formulations is adequate only when the bottom is close to planar. On more complex terrain, the effective footprint depends on slope, relief, and the relative orientation between the vehicle track and the local seabed surface. Consequently, uniform line spacing can produce substantial redundant overlap in some areas while still leaving under-covered regions elsewhere [3,4,6].

To avoid the under- and over-coverage induced by a fixed nominal spacing, coverage is evaluated from the local sensor–terrain geometry rather than from a prescribed constant swath width. Let ψ denote a candidate global line orientation and let s parameterize position along a survey line. For each sampled position s, the MBES fan is analyzed in the cross-track plane orthogonal to the vehicle heading. The boundary rays on the port and starboard sides are intersected with the prior bathymetric surface, and the first valid intersections determine the laterally reachable seafloor on each side of the vehicle.

This construction yields side-specific cross-track extents,

$$w_{\text{port}}(s, \psi) \quad \text{and} \quad w_{\text{star}}(s, \psi),$$

from which the effective local swath width is defined as

$$w(s, \psi) = w_{\text{port}}(s, \psi) + w_{\text{star}}(s, \psi).$$

Evaluating the two sides separately is necessary because local relief can make the insonified footprint strongly asymmetric. As a result, the sensor model returns a spatially varying swath profile along each candidate line, rather than a single nominal width applied uniformly over the scene.

Within the CPP formalism, this terrain-dependent sensing model defines the coverage operator used to score a candidate lawnmower layout. For each survey line, the operator maps the line and the prior bathymetric surface to a predicted covered seabed subset. Coverage completeness is then computed from the union of these subsets, while redundant sensing is quantified from their pairwise and higher-order intersections. This representation links line placement directly to the objective terms introduced in the problem setting: reducing route length and excess overlap without allowing uncovered gaps.

The use of a geometry-aware operator is motivated by the same limitation that motivates the planning problem itself. Because MBES swath width changes with local slope and relief, a constant-spacing approximation can be inconsistent with the actual footprint implied by the terrain. Methods based on idealized fan-shaped or circular sensing abstractions do not capture this coupling between seafloor geometry and sonar footprint, and are therefore less suitable for evaluating fixed-track survey layouts when a prior terrain model is available [26,31]. Likewise, recent adaptive-spacing studies support the need to vary spacing in nonuniform terrain rather than treating coverage width as constant [7,22].

In this work, the operator is used offline during mission design. The method therefore evaluates a nominal prior bathymetric model, nominal vehicle pose along each line, and declared sensor configuration. If the realized terrain differs from the prior map, if navigation error shifts the vehicle relative to the planned track, or if altitude deviations alter the effective footprint, then realized coverage can depart from the predicted coverage used by the optimizer. The model should therefore be read as a terrain-aware geometric swath approximation. It does not model beam-level bathymetric uncertainty, sound-speed refraction, roll/pitch/heave-dependent footprint deformation, closed-loop altitude control, or seabed backscatter quality. Predicted coverage is consequently a planning-layer metric rather than a hydrographic-quality assurance statement. We test part of this gap later with a Monte Carlo execution-uncertainty replay, while leaving online replanning, state-estimation feedback, and formal coverage assurance to execution-aware planners.

We model the MBES by its total opening angle φ and evaluate candidate coverage in the vertical plane orthogonal to the survey heading. This construction isolates the cross-track geometry that determines swath width for a fixed lawnmower pattern. Let D_0 be the depth at a reference point on the line, let a denote the local seabed slope magnitude, and let β denote the angle between the survey heading and the direction of steepest descent. Because along-track slope does not change the cross-track footprint in this local model, only the cross-track component of the terrain inclination is retained.

Define the effective cross-track inclination angle a_1 by projecting the local slope onto the direction normal to the survey line:

$$\tan a_1 = \tan a \cdot \cos(\beta - 90^\circ). \tag{1}$$

Under the local planar approximation, the bathymetry in the cross-track section is then represented as a linear depth profile,

$$D(x) = D_0 - x \tan(a_1), \tag{2}$$

where x is the horizontal offset measured from the reference point in the direction perpendicular to the survey line. This approximation provides the terrain-dependent depth needed to evaluate how swath width changes with survey orientation and local seabed geometry.

The local swath width is computed from the same cross-track geometry. For a survey line at position x, the outer port and starboard beams leave the transducer at half-angle $\varphi/2$. When the seafloor is laterally inclined, these two beams intersect the bottom at different slant ranges, which makes the upslope and downslope contributions to the insonified strip unequal. The horizontal swath width is therefore

$$W(x) = \left(\frac{D(x)\sin(\varphi/2)}{\sin(90^\circ + a_1 - \varphi/2)} + \frac{D(x)\sin(\varphi/2)}{\sin(90^\circ - a_1 - \varphi/2)} \right) \cos(a_1). \tag{3}$$

This expression explicitly captures the asymmetry induced by the local cross-track slope. When $a_1 = 0$, it simplifies to the flat-bottom relation $W(x) = 2D(x)\tan(\varphi/2)$.

In the planner, (2)–(3) define the terrain-dependent sensing model used to score candidate line sets under the fixed lawnmower traversal. For each candidate orientation, the model is queried repeatedly to estimate admissible line spacing and the resulting overlap or under-coverage along neighboring lines. These quantities are then passed to the subsequent orientation search and line-placement optimization stages. The notation used in this model is summarized in Table 2.

The model assumes that each local cross-track section can be approximated as planar. This approximation is appropriate for offline planning with a prior bathymetric surface, but its accuracy degrades where the terrain exhibits strong within-swath curvature, sharp ridges, or rapidly varying small-scale relief. In such regions, the predicted width should be interpreted as a local approximation rather than an exact footprint.

Table 2. Notation used by the terrain–sensor model and the adjacent-swath overlap calculation.

Symbol	Definition
D_0	Water depth at the local reference point (m)
α	Dominant seafloor slope angle relative to the horizontal
β	Angle between the survey-track direction and the direction of steepest descent
α_1	Effective cross-track slope angle in the plane normal to the survey track
x	Horizontal distance from the reference point measured in the cross-track direction (m)
$D(x)$	Water depth at position x (m)
ψ	Global survey-line orientation (°)
φ	Total multibeam opening angle (°)
$W(x)$	Predicted swath width at position x (m)
d	Spacing between adjacent survey lines (m)
η	Overlap ratio between adjacent swaths

We formulate offline survey design as a search over the global line orientation ψ and the sorted cross-track line positions $\mathbf{p} = [p_1, \ldots, p_N]$ of a fixed lawnmower pattern. Given $(\psi, \mathbf{p})$, the induced plan consists of N parallel survey lines traversed in alternating directions. The geometric route cost is the total traversal length

$$L(\psi, \mathbf{p}) = \sum_{i=1}^{N} \ell_i + \sum_{i=1}^{N-1} c_{i,i+1}, \tag{4}$$

where ℓ_i is the in-region length of line i and $c_{i,i+1}$ is the transition distance between consecutive lines under the prescribed sweep order. This objective reflects the two dominant contributors to mission effort in the present benchmark: productive survey distance along the lines and the regular turn-to-turn transit imposed by the parallel-line layout. The objective optimizes line spacing and cross-track placement rather than vehicle dynamics. Because the output remains a fixed parallel-line family, the line-change segments can be smoothed after planning with standard Dubins, clothoid, or spline transitions. A turning-aware post-evaluation can be written as $L_R(\psi, \mathbf{p}) = L(\psi, \mathbf{p}) + (N-1)\pi R_{\min}$ for a conservative semicircular line-change penalty, and a future execution-aware version can place $R_{\min}$, turn-time, or curvature penalties directly in the GA fitness.

Coverage is evaluated from the union of line-specific footprints predicted by the sensor–terrain model. Let Ω denote the target survey region and let $\mathcal{F}_i$ be the predicted footprint of line i. The predicted coverage rate is

$$CR(\psi, \mathbf{p}) = \frac{\left|\Omega \cap \bigcup_{i=1}^{N} \mathcal{F}_i\right|}{|\Omega|} \times 100\%. \tag{5}$$

The local overlap ratio between adjacent lines is still defined from the terrain-aware swath model as

$$\eta_i(x) = 1 - \frac{d_i}{W_i(x)}, \tag{6}$$

where $d_i = p_{i+1} - p_i$ and $W_i(x)$ is the effective swath width available at location x for that adjacent-line pair. Rather than enforcing a pointwise hard band everywhere, the implementation reports only the mean excess-overlap violation above the 20 percent ceiling,

$$O_{\text{ex}}(\psi, \mathbf{p}) = \frac{1}{|\Omega|} \int_{\Omega} \max(\eta(x) - 0.20,\, 0)\, d\Omega \times 100\%. \tag{7}$$

The target-overlap design margin is 15 percent, while the reported feasibility test accepts a plan only if

$$CR(\psi, \mathbf{p}) \geq 97.0\%, \qquad O_{\text{ex}}(\psi, \mathbf{p}) \leq 3.0\%. \tag{8}$$

This distinction is important for interpreting the benchmark. The 10–20 percent overlap interval is the design intent used when constructing candidate line families, whereas the reported acceptance rule is expressed through domain-level predicted coverage and mean excess-overlap violation.

Candidate plans are ranked by the same scalar score used in the implementation,

$$S(\psi, \mathbf{p}) = L(\psi, \mathbf{p}) + 80 \max(97.0 - CR(\psi, \mathbf{p}),\, 0) + 3\, O_{\text{ex}}(\psi, \mathbf{p}), \tag{9}$$

so under-coverage is penalized more strongly than moderate path-length differences, and residual overlap is treated as a secondary cost within the feasible region. The 97 percent coverage value is a benchmark acceptance threshold used to compare all methods under the same evaluator; it is not presented as a hydrographic survey standard or a field-coverage guarantee. This formulation targets offline survey design with a prior bathymetric model. If the realized seabed or executed trajectory deviates materially from the assumed prior and nominal line tracking, then actual overlap and coverage can depart from the predicted values used by the optimizer.

The implemented benchmark compares five method variants built on the same score and sensor–terrain evaluator: Fixed-Spacing, Simple Greedy, Adaptive Spacing without GA, Fixed-Swath GA, and Full Geometry-Aware Hybrid GA. The variants differ only in how the orientation and line positions are generated.

Discrete orientation scan and constant-spacing baseline

The first deterministic component scans candidate headings

$$\psi \in \{0^\circ, 15^\circ, \ldots, 165^\circ\}.$$

For each heading, a constant-spacing baseline is constructed from the scene-wide swath-width distribution. Let $\widetilde{W}_q(\psi)$ denote the q-quantile of the terrain-aware swath widths evaluated under heading ψ. For each

$$q \in \{0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50\},$$

a constant inter-line spacing is proposed as

$$d_q(\psi) = \widetilde{W}_q(\psi)(1 - \eta_{\text{tar}}), \qquad \eta_{\text{tar}} = 0.15. \tag{10}$$

The resulting parallel-line family is scored by (9), and the best candidate over all headings and constant-spacing quantiles defines the Simple Greedy baseline. Fixed-Spacing uses the same constant-spacing construction at the reference heading $\psi = 0^\circ$ only. This scan is not a proof of global optimality, but it is fully reproducible and provides a stable geometry-aware orientation prior for the later refinements.

Adaptive spacing without GA

The second deterministic component replaces a single scene-wide spacing by a cross-track spacing profile derived from local swath statistics. For each candidate heading, the rotated cross-track coordinate v is divided into 160 bins. In each bin, the planner computes a swath-profile quantile

$$\widehat{W}_q(v), \qquad q \in \{0.20, 0.25, 0.30, 0.35\},$$

and interpolates it over the full cross-track extent. Starting from the first admissible line position, the line family is then generated recursively by

$$p_{k+1} = p_k + \widehat{W}_q(p_k)(1 - \eta_{\text{tar}}). \tag{11}$$

The best heading/profile pair under (9) defines the Adaptive Spacing without GA baseline. This step is crucial for interpreting the results because it isolates how much benefit comes from terrain-aware spacing itself before any stochastic search is added.

GA refinement of line positions

The two GA variants keep the heading fixed and refine only the sorted cross-track positions $\mathbf{p}$. Fixed-Swath GA starts from the best constant-spacing candidate returned by the discrete orientation scan, whereas Full Geometry-Aware Hybrid GA starts from the adaptive-spacing baseline. In both cases, the number of lines is inherited from the deterministic base layout and remains fixed within a run.

The GA is real coded. The initial population contains the base layout plus nine jittered variants, where each gene is perturbed by zero-mean Gaussian noise with standard deviation 0.10 times the nominal median spacing. Fitness is evaluated on a stride-3 subsampled grid for speed. Tournament selection uses three candidates, arithmetic crossover draws $\alpha \sim \mathcal{U}(0.30, 0.70)$ and forms the child as $\alpha\mathbf{p}^{(a)} + (1 - \alpha)\mathbf{p}^{(b)}$, and mutation perturbs each gene independently with probability 0.30 using Gaussian noise with standard deviation 0.05 times the nominal spacing. A spacing-regularity penalty,

$$P_{\text{sep}}(\mathbf{p}) = 0.04 \sum_{i=1}^{N-1} \max\big(0,\, 0.25\,\bar{d} - (p_{i+1} - p_i)\big), \tag{12}$$

is added when adjacent lines collapse below one quarter of the nominal spacing $\bar{d}$. The final GA fitness is therefore

$$S_{\text{GA}}(\psi, \mathbf{p}) = S(\psi, \mathbf{p}) + P_{\text{sep}}(\mathbf{p}). \tag{13}$$

Each run uses 10 generations, population size 10, and elitism, so the best individual found so far is copied directly into the next generation. This deliberately small GA budget is appropriate because the discrete heading scan and adaptive-spacing stage already provide a strong base layout; the GA is a local refinement step around that layout rather than a blind global optimizer over arbitrary trajectories. The small population and short schedule therefore test whether residual overlap can be cleaned up with minimal additional search, while preserving offline planning efficiency. On the two GEBCO public scenes, the reported Hybrid GA planning times remain in the 0.32–0.69 s range, which is consistent with the planner's role as a rapid pre-mission line-layout tool. The output is a fixed-pattern line family with the same global heading as the base layout but cleaner cross-track placement under the terrain-aware evaluator.

Taken together, the optimizer preserves a fixed-pattern line-layout boundary. It tests whether prior-bathymetry-guided spacing and position refinement reduce unnecessary track length and overlap, relative to uniform spacing under the same traversal pattern, while maintaining predicted coverage. Route-order optimization, online replanning, and closed-loop adaptation to prior-map mismatch are treated as downstream extensions rather than hidden components of the present method.

For reproducibility, each benchmark run follows the same sequence: compute terrain-aware swath statistics for every candidate heading, generate the best deterministic baseline under (9), inherit that baseline into the relevant GA variant when stochastic refinement is enabled, and finally rescore the selected layout on the full evaluator grid before reporting path length, predicted coverage, excess overlap, planning time, and feasibility. Table 3 consolidates the fixed implementation settings used by the planner in the main benchmark.

Table 3. Fixed implementation settings used by the planner in the main benchmark. Sensitivity diagnostics change only the declared setting under test unless noted otherwise.

Component	Setting
Heading scan	$\psi \in \{0^\circ, 15^\circ, \ldots, 165^\circ\}$
Constant-spacing candidates	$q \in \{0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50\}$ with $d_q(\psi) = \widetilde{W}_q(\psi)(1 - \eta_{\text{tar}})$
Adaptive profile candidates	$q \in \{0.20, 0.25, 0.30, 0.35\}$ over 160 cross-track bins, followed by the recursive update in (11)
Target overlap margin	$\eta_{\text{tar}} = 0.15$
Acceptance rule	$CR(\psi, \mathbf{p}) \geq 97.0\%$ and $O_{\text{ex}}(\psi, \mathbf{p}) \leq 3.0\%$
Base GA population	Deterministic base layout plus nine jittered layouts
Initialization jitter	Zero-mean Gaussian noise with standard deviation 0.10 times the nominal median spacing
Selection and crossover	Tournament size 3; arithmetic crossover with $\alpha \sim \mathcal{U}(0.30, 0.70)$
Mutation	Per-gene probability 0.30; Gaussian perturbation with standard deviation 0.05 times the nominal spacing
Spacing-regularity penalty	Activated when adjacent lines fall below $0.25\,\bar{d}$
GA schedule	10 generations, population size 10, elitist carryover of the current best individual; used as local refinement after deterministic orientation and spacing initialization
Fitness evaluation	Stride-3 subsampled grid during GA; full evaluator grid for the reported metrics
Stochastic repeats	20 seeds for Fixed-Swath GA and Full Geometry-Aware Hybrid GA in the main benchmark

3.1. Experimental evaluation protocol

The numerical evaluation uses the benchmark suite reported in this study. The primary benchmark set consists of two public GEBCO 2025 bathymetry subsets, GEBCO Cascadia Margin and GEBCO Monterey Canyon, which are treated as publicly released gridded bathymetry inputs for an offline planning benchmark. The secondary benchmark set consists of three synthetic terrains, Flat Seafloor, Uniform Slope, and Complex Terrain, used to isolate mechanism and stress-test the fixed-heading parallel-line assumption. To reduce scene-selection risk without redefining the main benchmark, a supplemental four-window GEBCO expansion further probes Mariana Trench, Puerto Rico Trench, Mid-Atlantic Ridge, and Hawaii Ridge subsets with the same evaluator and five Hybrid GA seeds.

Five methods are compared under the same sensor–terrain evaluator: Fixed-Spacing, Simple Greedy, Adaptive Spacing without GA, Fixed-Swath GA, and Full Geometry-Aware Hybrid GA. The reported metrics are total path length, predicted coverage, excess overlap above the prescribed ceiling of 20 percent, planning time, and a feasibility flag. Lower values are preferred for path length, excess overlap, and computation time, whereas higher values are preferred for predicted coverage. The benchmark uses a 97.0% coverage target, a 15 percent target-overlap design margin, a 3.0 percent mean excess-overlap feasibility ceiling, a 120° MBES opening angle, candidate headings from 0° to 165° in 15° increments, 10 GA generations, a population size of 10, and 20 seeds for stochastic GA methods. Fixed-Spacing, Simple Greedy, and Adaptive Spacing without GA are deterministic in this implementation. The public windows are regional rather than sortie-scale, spanning roughly 128 × 178 km and 90 × 111 km after cropping, so absolute route lengths should be interpreted as large-area benchmark totals rather than single-mission AUV budgets.

The comparator set is chosen to isolate fixed-pattern line-layout effects rather than to rank all underwater CPP algorithms. Online bathymetric SLAM planners, cooperative multi-AUV planners, and energy-aware vehicle-dynamics planners solve broader problems with additional objectives, route topologies, or information states. Directly comparing against them would mix terrain-aware MBES spacing with route sequencing, localization, vehicle allocation, or control design. The baselines here instead form a controlled ladder inside the same fixed-pattern family: constant spacing, greedy spacing, adaptive spacing without stochastic refinement, GA under a fixed swath abstraction, and the full geometry-aware hybrid planner. This makes the ablation interpretable: any improvement can be attributed to terrain-dependent spacing, heading selection, or GA refinement rather than to a change in the mission-planning problem itself.

Table 4 records the provenance of the primary public benchmark set. Both public scenes are GEBCO 2025 grid subsets with complete valid-cell coverage in the saved manifests. They are therefore publicly released gridded bathymetry inputs to a numerical planner, not AUV mission logs, raw MBES returns, or field-executed survey tracks.

Table 4. Public GEBCO bathymetry subsets used as the primary external-data benchmark. Bounds are geographic crop limits in EPSG:4326; extents are computed from the locally projected evaluator grids used in this study. The listed grid resolutions are projected cell sizes after crop-and-project preprocessing, not the native angular spacing of the global GEBCO product.

Scene	Crop bounds	Extent	Grid resolution	Depth range	Valid
GEBCO Cascadia Margin	lon −126.8 to −125.2; lat 43.2 to 44.8	128.1 × 178.1 km	1195.383 m	1009–3101 m	1.0
GEBCO Monterey Canyon	lon −123.3 to −122.3; lat 35.3 to 36.3	90.3 × 111.3 km	758.720 m	892–3982 m	1.0

The use of GEBCO also imposes a data-level boundary. The official GEBCO documentation describes the GEBCO 2025 Grid as a global terrain model and an information product derived from heterogeneous source data and interpolation; it is accompanied by a Type Identifier (TID) grid that records source-data categories for grid cells [1]. The present benchmark uses GEBCO elevation subsets processed into locally projected evaluator grids, but it does not yet condition the planner on the TID layer or on source-specific uncertainty. Consequently, the public scenes should be interpreted as external public-grid benchmarks for offline line-layout evaluation, not as survey-grade terrain products or safety-of-navigation inputs.

In addition to the main benchmark suite, we run lightweight public-scene sensitivity diagnostics on declared sensor and planning inputs. The MBES-opening diagnostic reruns Fixed-Spacing, Adaptive Spacing without GA, and Full Geometry-Aware Hybrid GA on the two GEBCO scenes with opening angles of 100°, 110°, 120°, and 130°. The target-overlap diagnostic reruns the same methods at margins of 10%, 15%, and 20%, using the reported 120° MBES opening angle. The resolution diagnostic reruns the same methods on the native public grids and on 2× and 3× strided downsampled versions. The supplementary prior-map diagnostics rerun the public scenes under uniform depth biases of ±150 m and relief-scale factors of 0.7, 1.0, and 1.3, and a penalty-weight diagnostic reruns them across a 3×3 coverage/overlap objective-weight grid. A separate supplemental GEBCO four-window expansion checks whether the same low-overlap public-scene pattern persists on four additional public bathymetry windows. A separate USGS coarse-prior/fine-grid replay plans the selected public crops on 120, 300, and 600 m priors and then rescores the same line families on a finer public grid without re-optimization. The hybrid planner is summarized over seeds 0–4 for these diagnostics. These runs are not treated as replacements for the main 20-seed benchmark; they are used only to test whether the public-scene conclusions are fragile to small declared changes in sensor aperture, spacing margin, public-grid resolution, simple global prior-map perturbations, objective weighting, scene selection, or controlled prior-resolution mismatch.

We also add a post-planning execution-uncertainty replay on the two public GEBCO layouts. The replay does not re-optimize the line family. Instead, it perturbs the already selected Fixed-Spacing, Adaptive Spacing without GA, and representative Hybrid GA layouts, then recomputes predicted coverage and excess overlap on the same public-grid evaluator. The moderate-noise setting uses independent line-level cross-track perturbations with standard deviation $0.05\bar{d}$, a global cross-track offset with standard deviation $0.02\bar{d}$, heading noise with standard deviation 0.5°, a 3 percent global footprint-scale perturbation, and 2 percent spatially correlated local footprint-scale noise, where $\bar{d}$ is the median planned inter-line spacing. The strong-noise setting doubles these levels to $0.10\bar{d}$, $0.04\bar{d}$, 1.0°, 6 percent, and 4 percent. Each non-nominal setting uses 300 Monte Carlo trials per scene and method. This experiment is a numerical uncertainty replay, not a substitute for mission logs, but it directly tests whether the selected public layouts retain coverage and overlap margin under plausible post-planning perturbations.

4. Results

This section reports the benchmark suite used in this study. The primary results use two public GEBCO 2025 bathymetry subsets [1], GEBCO Cascadia Margin and GEBCO Monterey Canyon, as the main public-bathymetry benchmark for the offline planner. The secondary results use three synthetic terrains as controlled stress tests that isolate flat, sloped, and strongly heterogeneous seafloor structure, so that mechanism and failure mode can be separated from public-data realism. Supplemental diagnostics add four more GEBCO windows, a higher-resolution USGS public-grid extension, prior-resolution replay, objective-weight sensitivity, and execution-noise replay without merging those checks into the main 20-seed public averages. Only outputs generated from the reported terrain-aware MBES evaluator are included.

The same metrics and acceptance criteria defined in the protocol are used throughout this section. Coverage values are predicted coverage rates under the reported sensor–terrain model, not measured field coverage. The public windows are regional rather than sortie-scale, so the absolute route-length values should be interpreted as benchmark totals rather than as single-mission AUV budgets.

For readability, the Results section proceeds in seven linked steps. It first presents the two public GEBCO scenes as the main public-bathymetry benchmark, then extends the comparison to the full benchmark and to a scene-level overlap-regime diagnostic that explains why gain varies across scenes, then isolates the public-scene ablation and 20-seed repeatability of the stochastic stage, then reports sensitivity to declared sensor aperture, overlap margin, simplified prior-map perturbations, objective weighting, and public-grid resolution, then adds an execution-uncertainty replay on the public layouts, then adds a separate higher-resolution public-grid extension check, and finally tests coarse-prior/fine-grid replay on the same public USGS source. A supplemental four-window GEBCO expansion is reported as a scene-selection risk check rather than as a replacement for the primary two-scene benchmark.

Table 5 summarizes how the subsequent figures and tables are organized. Each figure or table is tied to one question and one supported claim so that the public-bathymetry and numerical-benchmark boundaries remain explicit.

Table 5. Results roadmap for the benchmark evidence. Each statement is limited to the evidence source that supports it and to the boundary that prevents over-interpretation.

Question addressed	Evidence used	Supported finding	Boundary
Does terrain-aware spacing help on public bathymetry grids?	Two GEBCO 2025 public subsets; Figures 3–4 and Table 6.	Hybrid GA reaches 98.97–99.63 percent predicted coverage, reduces mean scene-level excess-overlap violation from about 0.81 percent to 0.095 percent, and shortens mean scene-level path length by 0.75 percent.	Public-grid evidence for pre-mission AUV line planning.
Why are the public route-length gains modest while the stress-scene gains are large?	Scene-level baseline-overlap diagnostic spanning the main benchmark and the separate USGS extension; Figure 7.	Hybrid benefit is smallest when Fixed-Spacing already starts below about 2 percent excess overlap and largest when the Fixed baseline enters the 28–30 percent overlap-stress regime.	Descriptive diagnostic from the same numerical evaluator; it explains this benchmark but is not a universal field law.
Is GA the main source of the public-scene gain?	Fixed-Spacing vs Adaptive Spacing without GA vs Hybrid GA ablation, plus an equal-budget PSO local-refinement check; Figures 8 and 9.	Adaptive spacing captures nearly all public route shortening; GA mainly cleans residual overlap and stabilizes the final layout.	No claim of global optimality or GA-only innovation.
Where does the fixed-pattern assumption fail?	Synthetic complex-relief stress test and public sensitivity diagnostics; Figures 5–6, Table 11, Figures 10–11, and the supplementary prior-map perturbation checks.	Complex Terrain remains infeasible at 96.82 percent coverage; Monterey becomes marginal under 3× grid coarsening; simple global prior perturbations and the tested MBES opening-angle range do not move the public-scene layouts.	Single-heading lawnmower design is parameter- and resolution-sensitive, but not driven by trivial global prior offsets.
Do the selected public layouts retain margin under execution noise?	Post-planning execution-uncertainty replay with 300 Monte Carlo trials per non-nominal setting; Figure 12.	Under moderate perturbations, Hybrid GA remains feasible in 299/300 Cascadia trials and 300/300 Monterey trials; under stronger perturbations, all methods lose substantial feasibility margin.	Numerical perturbation replay only; not a mission-log or control-system validation.
Does the pattern persist on a higher-resolution public grid?	Independent USGS Southern Cascadia 30 m public-grid extension; Figure 13.	The high-complexity crop changes from infeasible Fixed-Spacing to feasible terrain-aware layouts, with Hybrid GA reaching 98.44 percent coverage and 1.73 percent excess overlap.	Separate transfer check; not merged into the main GEBCO benchmark averages.
Does the line family survive prior-resolution mismatch?	Coarse-prior/fine-grid replay on the USGS public grid; Figure 14.	On the high-complexity crop, Hybrid GA remains feasible when planned from 120–600 m priors and replayed on the fine grid, with 97.91–98.31 percent coverage and 1.31–2.16 percent excess overlap.	Public-grid replay only; not mission-log execution or survey-grade uncertainty certification.

Figure 2 summarizes the five benchmark scenes used in the main benchmark configuration. The synthetic scenes provide a clean mechanism study, whereas the two GEBCO subsets provide the main public-bathymetry test of the planner on non-synthetic seabed structure. The benchmark assumes that the prior bathymetric surface is available at planning time, so the figures and tables below should be read as public-data evidence for geometry-aware pre-mission line design, with survey-log and sea-trial transfer left to subsequent validation.

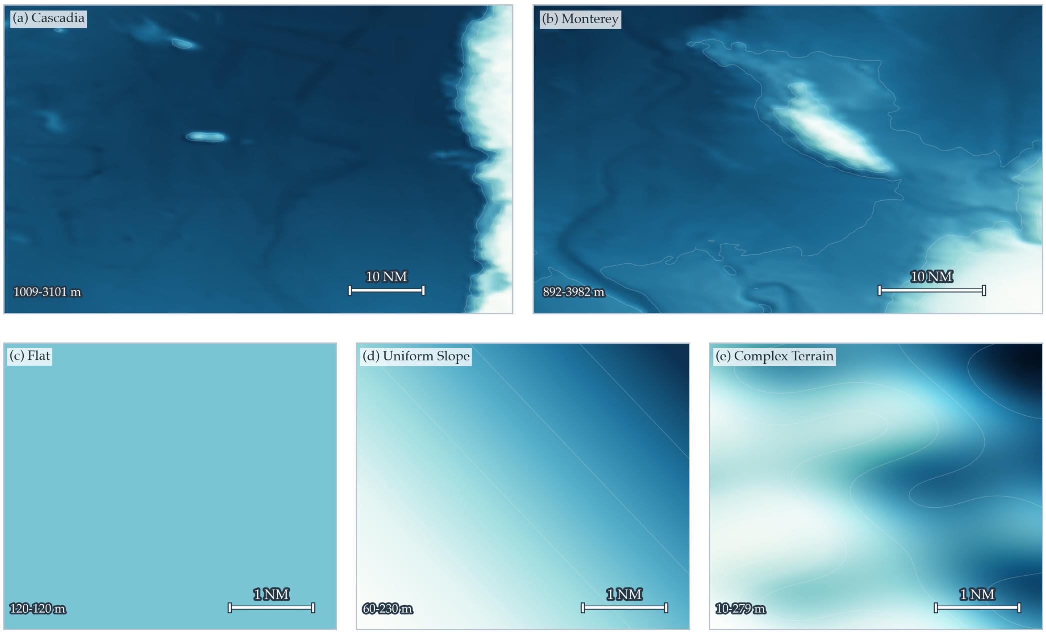

Figure 2. Benchmark atlas for the five terrains used in this study. Panels (a) and (b) are public GEBCO 2025 bathymetry subsets used as the main external-data benchmark; panels (c)–(e) are synthetic stress-test terrains used to isolate flat, sloped, and complex-relief mechanisms. Each panel preserves its own depth range and scale because the public and synthetic scenes differ substantially in physical extent and relief.

4.1. Public-scene External-Data Evidence

Figures 3 and 4 present the two public results as scene cards with three method-specific zoom panels on a shared terrain window, a full-scene context map that marks the zoom location, a compact metric table, and a small improvement panel. The zoom panels highlight local line-placement differences, while the tabulated metrics remain full-scene values. This layout reduces overplotting, removes the unused map regions that would dominate a full-scene strip layout, and makes it easier to separate genuine geometry change from small metric cleanup. On GEBCO Cascadia Margin, the fixed-spacing baseline and the two terrain-aware planners all retain a horizontal sweep family, and the difference between Adaptive Spacing without GA and the Full Geometry-Aware Hybrid GA is small. On GEBCO Monterey Canyon, terrain-aware adaptation rotates the survey family from the baseline's 0° heading to a 90° heading and reduces the line count from 73 to 59. This is the clearest public-scene example of the planner exploiting terrain geometry rather than merely tightening a fixed-spacing layout.

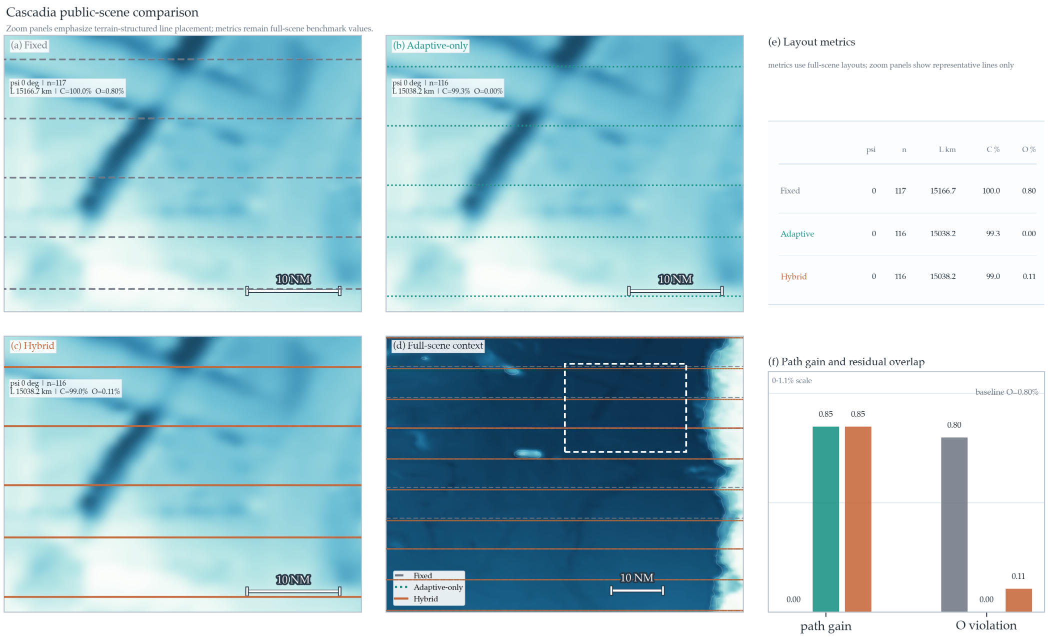

Figure 3. GEBCO Cascadia Margin public-scene comparison on the benchmark reported in this study. Panels (a)–(c) show Fixed-Spacing, Adaptive Spacing without GA, and Full Geometry-Aware Hybrid GA on a shared relief-focused zoom window, so the line-placement differences can be read without the large empty basin region dominating the figure. Panel (d) restores the full-scene context and marks the zoom window location. Panel (e) reports the heading, line count, path length L, predicted coverage C, and excess-overlap violation O computed from the full scene, while panel (f) converts those values into relative path gain and remaining overlap against Fixed-Spacing. Only representative lines are drawn for readability, but all reported metrics use the full layouts. All three methods retain the 0° sweep family in this control scene.

Cascadia functions as the moderate-relief control case for the public benchmark. The scene confirms that geometry-aware planning does not need a dramatic line-family rotation to be useful: even when all three methods retain the same horizontal sweep structure, terrain-aware spacing still removes the baseline overlap surplus and concentrates the remaining gain into cleaner line placement rather than into a visually different traversal pattern.

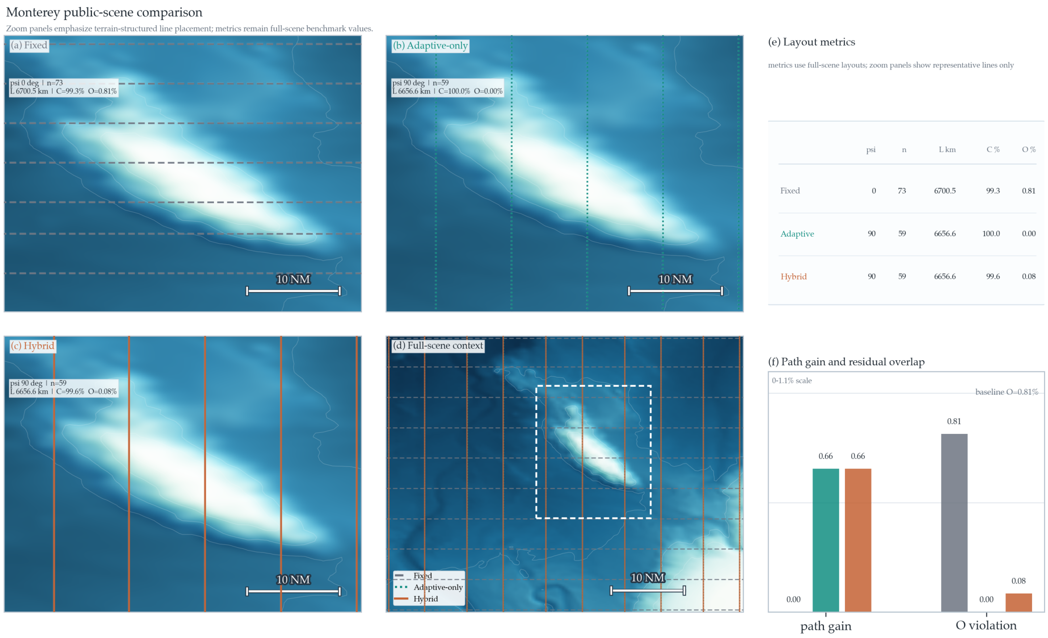

Figure 4. GEBCO Monterey Canyon public-scene comparison on the benchmark reported in this study. The panel structure and metric definitions match Figure 3: panels (a)–(c) are method-specific zooms on a shared canyon window, panel (d) restores the full-scene context and zoom location, panel (e) reports full-scene metrics, and panel (f) summarizes relative path gain and residual overlap against Fixed-Spacing. This is the strongest public-scene structural-change case: the Fixed-Spacing layout keeps the 0° family, whereas the Adaptive and Hybrid layouts rotate to 90° and reduce the line count from 73 to 59. The plotted line families are representative subsets for readability, whereas the reported metrics use the full-scene layouts.

Table 6 summarizes the public-scene comparison from the reported benchmark. On the two GEBCO scenes, the Full Geometry-Aware Hybrid GA achieved a 0.75 percent relative shortening of mean scene-level path length compared with Fixed-Spacing, reduced mean scene-level excess-overlap violation from about 0.81 percent to 0.095 percent, and maintained mean coverage of 99.30 percent. The main public result is therefore better overlap control and a cleaner fixed-line layout, rather than a large path-length reduction.

Table 6. Quantitative comparison on the two public GEBCO scenes from the reported benchmark. Fixed-Spacing and Adaptive Spacing without GA are deterministic single runs. Full Geometry-Aware Hybrid GA reports the mean over 20 seeds. Excess overlap is the area-averaged violation above the admissible overlap ceiling of 20 percent. Lower path length, excess overlap, and planning time are better; higher predicted coverage is better.

Scene	Method	Path Length (km)	Coverage (%)	Excess Overlap (%)	Planning Time (s)
GEBCO Cascadia Margin	Fixed-Spacing	15166.71	**100.00**	0.80	**0.07**
	Adaptive Spacing without GA	15038.22	99.33	**0.00**	0.75
	Full Geometry-Aware Hybrid GA	**15038.16**	98.97	0.11	0.69
GEBCO Monterey Canyon	Fixed-Spacing	6700.52	99.33	0.81	**0.04**
	Adaptive Spacing without GA	6656.62	**100.00**	**0.00**	1.89
	Full Geometry-Aware Hybrid GA	**6656.57**	99.63	0.08	0.32

Taken together, the two public scene cards reveal two distinct mechanisms. Cascadia is mostly an overlap-cleanup control case: the line family stays horizontal, and terrain-aware spacing mainly removes redundant overlap within an already regular sweep structure. Monterey is the stronger structural-change case: terrain-aware planning rotates the family to 90° and collapses the line count from 73 to 59. In both scenes, however, the gap between Adaptive Spacing without GA and the Full Hybrid GA remains very small in path length. The public-scene ablation therefore shifts the contribution claim in an important way: the public benefit is better overlap discipline and layout regularization under a fixed traversal, while the GA mainly regularizes the final solution, suppresses residual overlap, and improves seed-to-seed stability.

To make the vehicle-kinematics issue explicit without claiming field execution, Table 7 adds a post-planning turn-cost diagnostic for the two public layouts. The diagnostic uses the same reported line counts and geometric path lengths and evaluates the first-order effective length $L_R = L + (N-1)\pi R_{\min}$ for minimum-turn-radius values of 25, 50, and 100 m. This does not replace a Dubins- or controller-level feasibility study, but it checks whether the public-scene gain disappears under a conservative semicircular line-change penalty. On Cascadia, the turn-aware gains remain essentially unchanged because the terrain-aware layouts reduce the line count by only one. On Monterey, the terrain-aware layouts use 59 rather than 73 lines, so including turn cost slightly increases the effective path-gain estimate.

Table 7. Turning-aware post-evaluation for the two public GEBCO line families. L_R uses the evaluation-only semicircular line-change penalty $L_R = L + (N-1)\pi R_{\min}$. Gains are relative to Fixed-Spacing under the same $R_{\min}$, and all entries are recomputed from `run_5/benchmark_method_statistics.csv`.

Scene	Method	Lines	Turns	L (km)	G_L (%)	$G_{R=25}$ (%)	$G_{R=50}$ (%)	$G_{R=100}$ (%)
GEBCO Cascadia Margin	Fixed-Spacing	117	116	15166.71	0.00	0.00	0.00	0.00
	Adaptive Spacing without GA	116	115	15038.22	0.85	0.85	0.85	0.85
	Full Geometry-Aware Hybrid GA	116	115	15038.16	0.85	0.85	0.85	0.85
GEBCO Monterey Canyon	Fixed-Spacing	73	72	6700.52	0.00	0.00	0.00	0.00
	Adaptive Spacing without GA	59	58	6656.62	0.66	0.67	0.69	0.72
	Full Geometry-Aware Hybrid GA	59	58	6656.57	0.66	0.67	0.69	0.72

A supplemental four-window GEBCO expansion further checks whether the two primary GEBCO windows were unusually favorable. The added Mariana Trench, Puerto Rico Trench, Mid-Atlantic Ridge, and Hawaii Ridge crops remain feasible for all tested methods under the same evaluator. Fixed-Spacing starts in the same low-overlap regime as the two primary GEBCO windows, with 0.44–0.68 percent excess overlap. Adaptive Spacing without GA removes this excess overlap and yields 0.49–0.68 percent path gain, while the five-seed Hybrid GA produces nearly the same path-gain range with residual excess overlap of 0.01–0.18 percent and feasible seed rate 1.0. This expansion does not prove geographic representativeness, but it makes the public-data story less dependent on only two hand-visible scenes: additional GEBCO windows also behave like low-overlap public-grid cases where terrain-aware spacing mainly improves overlap discipline rather than producing large route compression.

4.2. Cross-scene Mechanism and Failure Boundary

The synthetic stress tests clarify both why the public gains are limited and where the present method breaks. On Flat Seafloor, the Full Geometry-Aware Hybrid GA achieved a 3.56 percent relative route shortening compared with Fixed-Spacing and removed excess overlap while remaining feasible. On Uniform Slope, the path gain increased to 7.69 percent, with near-saturated predicted coverage (99.999 percent) and only 0.044 percent excess overlap. On the hardest Complex Terrain scene, the route shortened from 342.38 km to 254.13 km and excess overlap fell from 28.50 percent to 7.18 percent, but coverage remained at 96.82 percent and the feasibility flag stayed 0.0 across seeds. Adaptive Spacing without GA was almost identical on that scene, so the results reveal a genuine failure case rather than a solved benchmark. Figure 5 visualizes the failure rather than only reporting the aggregate metric: terrain-aware layouts move the zero-coverage cells away from the broad Fixed-Spacing gap pattern and suppress the large overlap field, but localized uncovered patches remain in the complex-relief scene. The most likely interpretation is that terrain-aware spacing is necessary but not sufficient once relief becomes too heterogeneous for a single-heading parallel-line family to represent the survey efficiently.

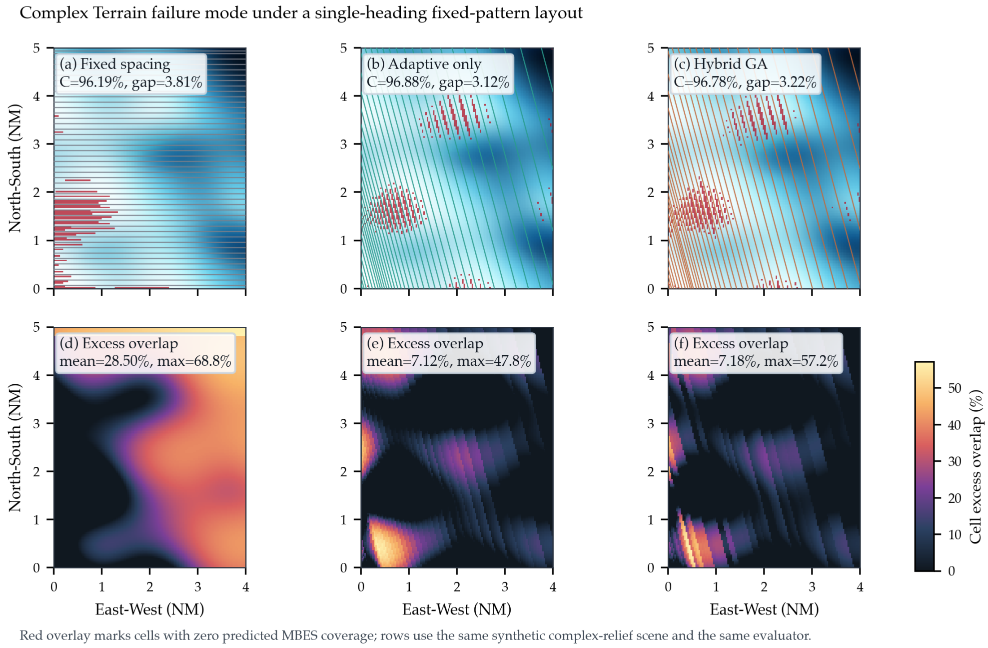

Figure 5. Failure-mode visualization for the synthetic Complex Terrain stress test. Panels (a)–(c) overlay zero-predicted-coverage cells in red on the terrain and planned line family for Fixed-Spacing, Adaptive Spacing without GA, and a representative Hybrid GA run. Panels (d)–(f) show the corresponding cellwise excess-overlap fields. The figure is recomputed from `run_5/complex_terrain_failure_mode_summary.csv` and the same sensor–terrain evaluator as the main benchmark. The result is a numerical boundary-of-validity diagnostic: adaptive spacing substantially reduces overlap but does not remove all predicted gaps under the present single-heading fixed-pattern assumption.

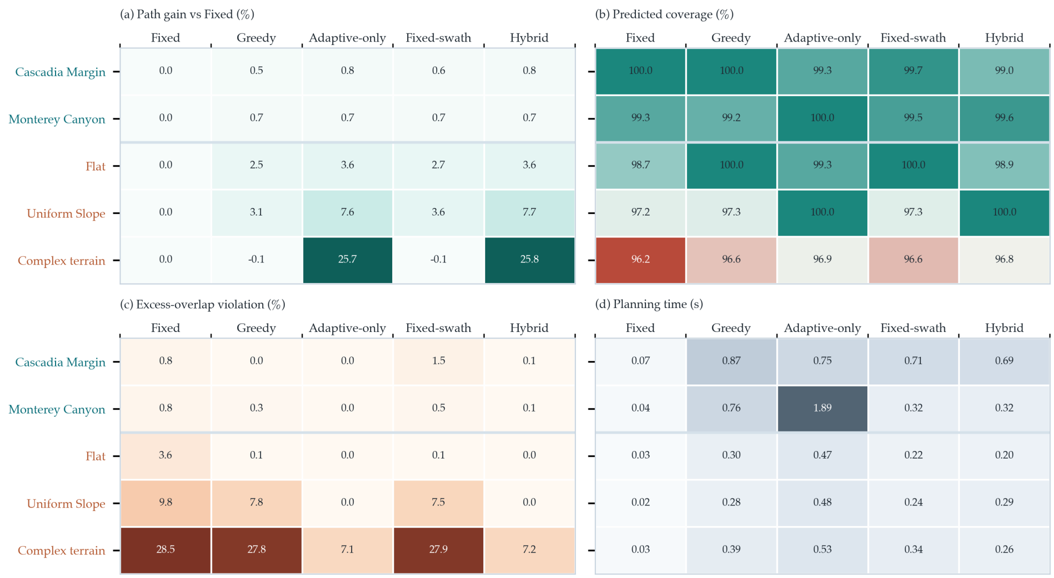

Figure 6. Cross-scene metric matrix from the reported benchmark. Each panel is a scene-by-method heatmap of benchmark means: path gain relative to Fixed-Spacing, predicted coverage, excess-overlap violation, and planning time. The top two rows are the public GEBCO scenes and the bottom three rows are synthetic stress tests, so the figure exposes both the public-data pattern and the complex-terrain failure boundary in one compact view. Stochastic GA methods report 20-seed means, and all coverage values are numerical predictions rather than field measurements.

Figure 6 then lifts the argument from the two public cards to the full benchmark. First, the route-gain panel makes the terrain-heterogeneity effect explicit: the adaptive-only and hybrid methods deliver much larger relative savings on the synthetic scenes than on the public GEBCO subsets. Second, the coverage-margin panel shows why the Complex Terrain result is a failure case rather than a hidden success: the large path gain is accompanied by coverage below the 97 percent threshold. Third, the overlap panel shows that Fixed-Swath GA is not enough by itself; its path gain can be competitive, but its excess-overlap violation remains much larger than that of terrain-aware spacing. Fourth, the planning-time panel confirms that all tested methods remain in the offline-design regime for these benchmark sizes. Table 8 provides the corresponding compact numerical matrix for readers who need the exact cross-scene values behind the visual summary.

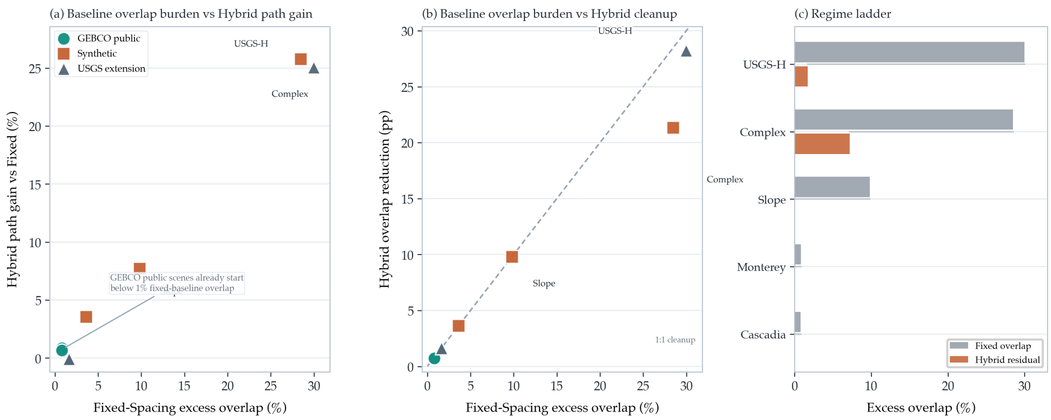

Figure 7. Scene-level regime diagnostic linking the Fixed-Spacing baseline to the observed Hybrid benefit. Each point is one scene from either the reported main benchmark or the separate USGS 30 m extension. Panel (a) plots Fixed-Spacing excess-overlap burden against Hybrid path gain relative to Fixed-Spacing. Panel (b) plots the same burden against Hybrid overlap reduction. Panel (c) gives a ranked overlap-regime ladder, comparing the Fixed-Spacing burden with the Hybrid residual on representative scenes. The GEBCO public scenes cluster in a low-overlap regime below 1 percent excess overlap, whereas Complex Terrain and the high-complexity USGS crop start near 30 percent excess overlap and therefore show the largest geometry-aware gains. The USGS points strengthen the mechanism interpretation but are not merged into the main benchmark averages reported elsewhere in the paper.

Figure 7 makes the cross-scene mechanism more specific. Across the five benchmark scenes and the three USGS extension crops, the dominant predictor of Hybrid benefit is not scene provenance by itself but the excess-overlap burden already present in the Fixed-Spacing baseline. The two GEBCO public scenes start at only 0.80 and 0.81 percent excess overlap and therefore yield only 0.66–0.85 percent route shortening, even though they still benefit from cleaner overlap control. The low- and medium-complexity USGS crops behave similarly: Fixed-Spacing already holds excess overlap at 1.633 percent, so Hybrid GA mainly removes that small surplus at negligible route-length cost. By contrast, Complex Terrain and the high-complexity USGS crop begin at 28.50 and 29.96 percent excess overlap and then deliver 25.77 and 25.04 percent path shortening, respectively, together with 21.32 and 28.23 percentage-point overlap cleanup. This diagnostic clarifies why the public gains are modest without weakening them: the reported GEBCO pair lives in a low-overlap regime rather than in the near-failure regime exposed by the harder synthetic scene and the high-complexity public-grid extension crop.

Table 8. Compact all-scene method matrix from the reported benchmark. Path gain is measured relative to Fixed-Spacing within the same scene. Coverage and excess overlap are reported in percent. Compact method labels abbreviate the five methods defined in the experimental protocol. Feasibility is the reported acceptance flag or, for stochastic methods, the mean feasibility rate over 20 seeds.

Scene	Method	Gain (%)	Coverage (%)	Excess Overlap (%)	Feas.
GEBCO Cascadia	Fixed	0.0	100.0	0.8	1.0
	Greedy	0.5	100.0	0.0	1.0
	Adaptive without GA	0.8	99.3	0.0	1.0
	Fixed-swath GA	0.6	99.7	1.5	1.0
	Hybrid GA	0.8	99.0	0.1	1.0
GEBCO Monterey	Fixed	0.0	99.3	0.8	1.0
	Greedy	0.7	99.2	0.3	1.0
	Adaptive without GA	0.7	100.0	0.0	1.0
	Fixed-swath GA	0.7	99.5	0.5	1.0
	Hybrid GA	0.7	99.6	0.1	1.0
Flat	Fixed	0.0	98.7	3.6	0.0
	Greedy	2.5	100.0	0.1	1.0
	Adaptive without GA	3.6	99.3	0.0	1.0
	Fixed-swath GA	2.7	100.0	0.1	1.0
	Hybrid GA	3.6	98.9	0.0	1.0
Uniform Slope	Fixed	0.0	97.2	9.8	0.0
	Greedy	3.1	97.3	7.8	0.0
	Adaptive without GA	7.6	100.0	0.0	1.0
	Fixed-swath GA	3.6	97.3	7.5	0.0
	Hybrid GA	7.7	100.0	0.0	1.0
Complex Terrain	Fixed	0.0	96.2	28.5	0.0
	Greedy	-0.1	96.6	27.8	0.0
	Adaptive without GA	25.7	96.9	7.1	0.0
	Fixed-swath GA	-0.1	96.6	27.9	0.0
	Hybrid GA	25.8	96.8	7.2	0.0

The planning-time panel shows that runtime is not the dominant weakness of the present implementation. On the current workstation, all public-scene planners remained within about two seconds per scene, with Fixed-Spacing fastest and the terrain-aware planners still comfortably in the offline-planning regime. The method therefore remains limited primarily by modeling scope rather than by brute-force computational latency.

4.3. Public-scene Ablation and Seed-level Repeatability

Figure 8 isolates the two pieces of evidence most relevant to the public-data contribution claim: the ablation that separates adaptive spacing from GA refinement, and the seed-level dispersion of the Hybrid GA on the two public scenes.

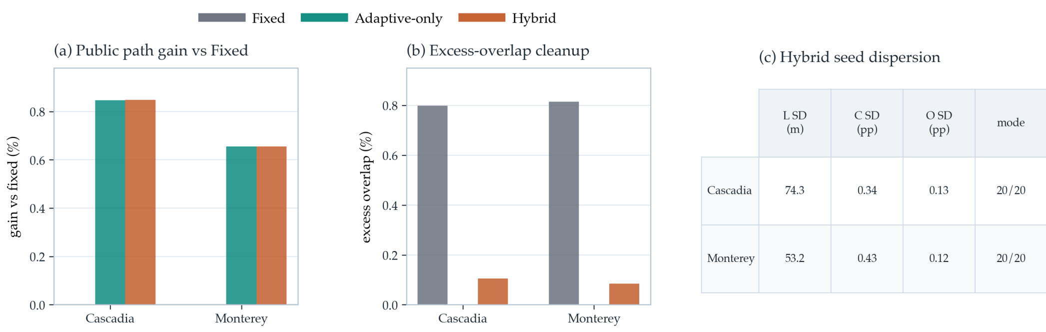

Figure 8. Public-scene ablation and seed-level dispersion from the reported benchmark. Panels (a) and (b) compare Fixed-Spacing, Adaptive Spacing without GA, and Full Geometry-Aware Hybrid GA on path gain relative to Fixed-Spacing and on excess-overlap violation for the two public GEBCO scenes. Panel (c) reports the standard deviation of Hybrid GA path length L, predicted coverage C, and excess overlap O across 20 seeds, together with the count of seeds that converged to the dominant orientation/line-count mode.

The main benchmark allows a direct 20-seed repeatability check on both public scenes. Table 9 shows that the Full Geometry-Aware Hybrid GA converged to the same heading mode and the same line-count mode in every seed: 0°/116 lines on GEBCO Cascadia Margin and 90°/59 lines on GEBCO Monterey Canyon. Path-length standard deviation stayed below 0.08 km, coverage standard deviation below 0.43 percentage points, and excess-overlap standard deviation below 0.13 percentage points. This is strong seed-level stability for the public-data benchmark, although it should not be confused with a full sensitivity characterization over hyperparameters, map quality, or execution uncertainty.

Table 9. Twenty-seed repeatability of the Full Geometry-Aware Hybrid GA on the two public GEBCO scenes from the reported benchmark. Heading and line-count modes were identical across all 20 seeds for each scene. Entries report mean ± standard deviation.

Scene	Heading Mode	Line Count Mode	Path Length (km)	Coverage (%)	Excess Overlap (%)
GEBCO Cascadia Margin	0°	116	15038.16 ± 0.07	98.97 ± 0.34	0.11 ± 0.13
GEBCO Monterey Canyon	90°	59	6656.57 ± 0.05	99.63 ± 0.43	0.08 ± 0.12

Table 10 adds a bootstrap confidence check over the same 20 Hybrid GA seeds. The deterministic Fixed-Spacing value is used only as the reference for gain and overlap cleanup; it is not treated as a random sample. The resulting intervals are intentionally reported as seed-level uncertainty, not as geographic or field-performance confidence intervals. They show that the modest public path gain is extremely stable under the present GA schedule, while the overlap-control conclusion remains positive even after accounting for seed-level dispersion.

Table 10. Seed-level bootstrap confidence intervals for the Full Geometry-Aware Hybrid GA on the public GEBCO scenes. Intervals are 95% bootstrap CIs of the seed mean from 20 Hybrid GA runs and are recomputed from `run_5/public_hybrid_bootstrap_ci.csv`. G_L is path gain relative to the deterministic Fixed-Spacing layout; ΔO is excess-overlap cleanup relative to Fixed-Spacing.

Scene	Feasible seeds	G_L mean [95% CI] (%)	Coverage mean [95% CI] (%)	Excess overlap mean [95% CI] (%)	ΔO mean [95% CI] (pp)
GEBCO Cascadia Margin	20/20	0.8476 [0.8474, 0.8478]	98.9667 [98.8333, 99.1000]	0.1055 [0.0561, 0.1645]	0.6930 [0.6345, 0.7425]
GEBCO Monterey Canyon	20/20	0.6558 [0.6555, 0.6562]	99.6250 [99.4583, 99.7917]	0.0849 [0.0410, 0.1380]	0.7300 [0.6746, 0.7741]

Taken together, the repeatability and bootstrap results show strong seed-level stability on the public-data benchmark. This observation is limited to the current configuration and does not amount to a full sensitivity analysis across GA settings, bathymetry resolution, spatially varying prior-map mismatch, or uncertainty margins.

4.4. External Optimizer Sanity Check

To check whether the public result depends narrowly on choosing GA as the local search operator, we added an equal-budget Particle Swarm Optimization (PSO) diagnostic on the two public GEBCO scenes. The PSO variant starts from the same Adaptive Spacing baseline as the Hybrid GA, keeps the same heading and line count, and refines only cross-track positions with 10 particles and 10 iterations. This makes it a like-for-like local-refinement sanity check rather than a new global CPP competitor. The diagnostic uses seeds 0–9 and does not replace the 20-seed Hybrid GA benchmark reported above.

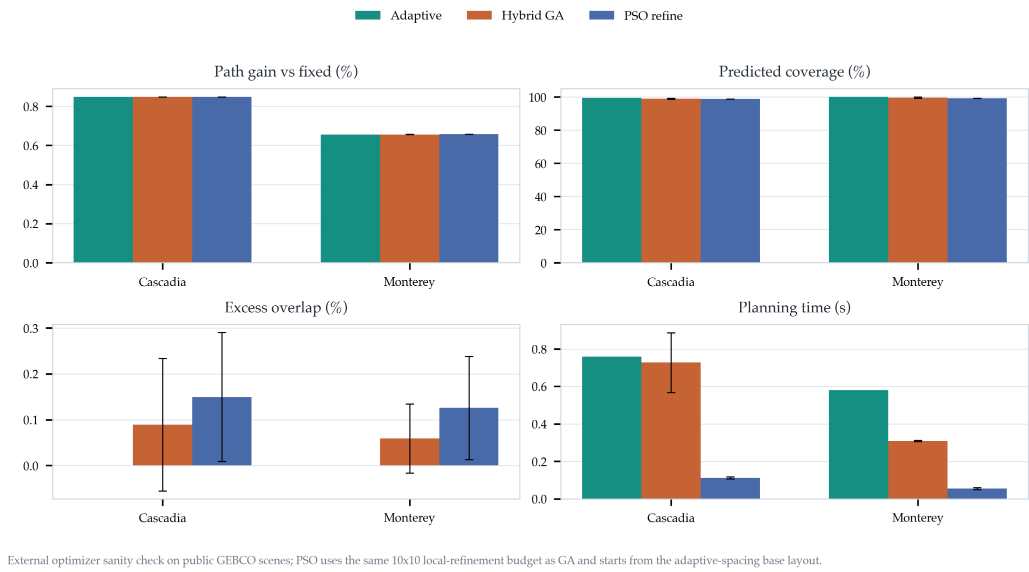

Figure 9. Equal-budget PSO local-refinement diagnostic on the two public GEBCO scenes. PSO starts from the same adaptive-spacing base layout as the Hybrid GA and uses the same 10-by-10 local-refinement budget. Bars report public-scene means, and error bars report seed-level standard deviations for the stochastic refiners. The diagnostic is included to test optimizer dependence; it is not a replacement for the main 20-seed Hybrid GA benchmark.

The PSO check supports the interpretation that terrain-aware spacing, not a particular metaheuristic label, is the main source of the public-scene path shortening. PSO produced comparable or slightly shorter path lengths, with 0.848 percent and 0.657 percent path gain on Cascadia and Monterey, respectively. However, it also produced lower predicted coverage than Hybrid GA in this diagnostic and higher residual excess overlap: 0.150 percent versus 0.089 percent on Cascadia, and 0.126 percent versus 0.059 percent on Monterey. The result therefore sharpens the claim rather than inflating it. The stochastic refinement step can be implemented by more than one local optimizer, but the reported Hybrid GA remains the more overlap-disciplined choice under the equal small-budget setting used here.

4.5. Sensitivity to Declared Sensor, Map, and Planning Inputs

Table 11 summarizes the additional public-scene diagnostics around the main benchmark. These diagnostics make the numerical claim harder to dismiss as a single-parameter artifact by testing sensor aperture, design margin, prior-map perturbation, public-grid resolution, and execution noise around the same reported layouts. Across 100°–130° MBES opening-angle reruns, the Hybrid GA retained the same public-scene heading and line-count modes as the main benchmark: 0°/116 lines on Cascadia and 90°/59 lines on Monterey, with all tested runs feasible in the five-seed diagnostic. This result should be interpreted as a local sensor-parameter check under the same idealized footprint model, not as a calibration study of real sonar performance.

Table 11. Public-scene sensitivity diagnostics around the main benchmark. Hybrid GA summaries use seeds 0–4 for the diagnostics, whereas the main reported benchmark uses 20 seeds.

Diagnostic	Values tested	Observed public-scene outcome	Remaining boundary
MBES opening angle	100°, 110°, 120°, 130°	Hybrid GA kept the same heading and line-count modes on both public scenes; five-seed feasibility stayed 1.0, with mean predicted coverage of 98.93% on Cascadia and 99.83% on Monterey.	Local idealized-footprint check, not sonar calibration or field-performance evidence.
Target-overlap margin	10%, 15%, 20%	All tested public layouts remained feasible, but selected headings and line counts changed across margins.	The margin is a declared design parameter, not a universal constant.
Public-grid resolution	native, 2×, 3× stride	Cascadia remained feasible across tested strides; Monterey Hybrid GA became marginal at 3× stride with mean coverage near 97.07% and feasibility rate 0.8.	Coarse public grids can erode evaluated coverage margin.
Simple prior-map perturbation	depth bias ±150 m; relief scale 0.7–1.3	Public-scene headings, line counts, and reported means were unchanged under these global perturbations.	Does not test spatially structured map error or navigation drift.
Objective penalty weights	coverage penalty 40, 80, 160; overlap penalty 1.5, 3.0, 6.0	Public-scene headings and line counts stayed unchanged across the 3×3 grid; Hybrid GA path-gain ranges were 0.8474–0.8476% on Cascadia and 0.6551–0.6574% on Monterey.	Objective-weight diagnostic only; it does not replace broader multi-objective tuning.

Figure 10 reports the public-scene target-overlap diagnostic. All tested settings remain feasible on both GEBCO scenes, so the main conclusion that terrain-aware spacing can control overlap while preserving predicted coverage is not limited to the 15 percent target-overlap value. The diagnostic also exposes a useful boundary: the selected layout is not invariant to this design margin. On Cascadia, the adaptive and hybrid planners switch between 90° layouts at the 10 percent and 20 percent settings and the 0° layout used in the main benchmark configuration at 15 percent. On Monterey, the 10 percent setting selects a 75° adaptive family, the 15 percent setting selects the main 90° family, and the 20 percent setting returns to a denser 0° family. This sensitivity is not a failure of the benchmark, but it means that the overlap margin should be treated as a declared planning parameter rather than as a universal constant.

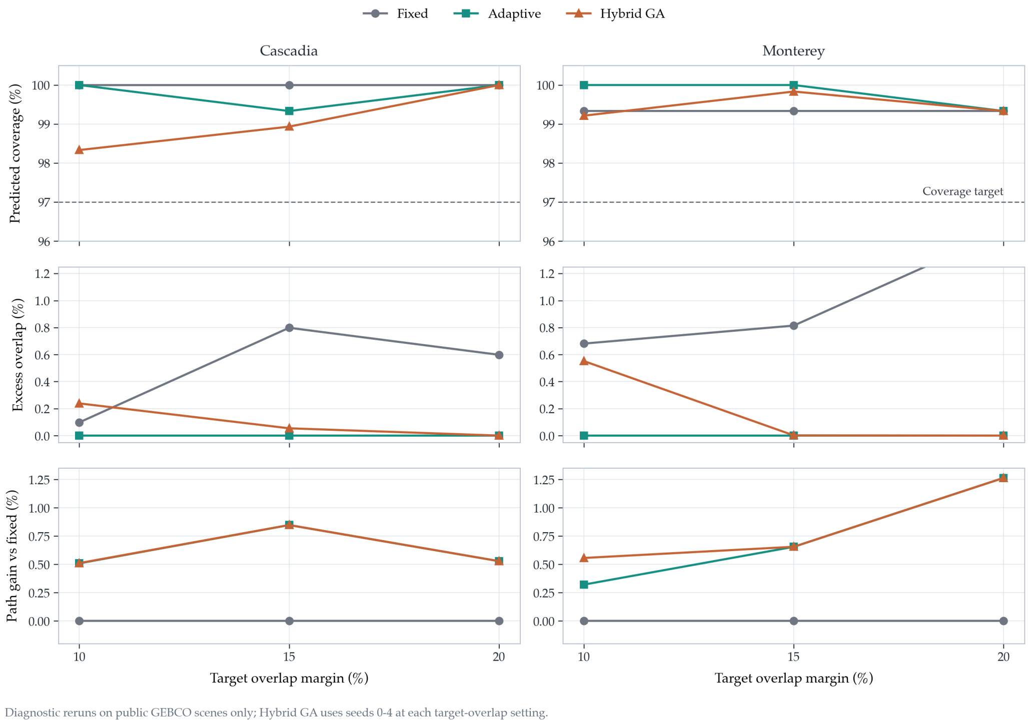

Figure 10. Public-scene target-overlap sensitivity diagnostic. Fixed-Spacing, Adaptive Spacing without GA, and Full Geometry-Aware Hybrid GA are rerun on the two GEBCO public scenes at target-overlap margins of 10, 15, and 20 percent using the same reported evaluator and a 120° MBES opening angle. Hybrid GA reports the mean over seeds 0–4 for this diagnostic; the main benchmark results remain the 20-seed values reported in this study. The figure shows that all tested public layouts remain feasible, but the selected line family and path gain depend on the declared design margin.

Figure 11 adds a second diagnostic for GEBCO grid resolution. On Cascadia, the terrain-aware line family remains feasible under native, 2×, and 3× strided grids, although coverage margin decreases from the native value as the grid is coarsened. Monterey is more sensitive. The adaptive-only planner remains feasible and holds predicted coverage at 100.0 percent in the downsampled diagnostic, but the Hybrid GA mean drops to 97.07 percent coverage at the 3× stride and the feasibility rate falls to 0.8 across seeds 0–4. This does not overturn the main native-grid result; it shows that the public benchmark is resolution dependent and that coarse bathymetric products can reduce the safety margin of the evaluated fixed-line family.

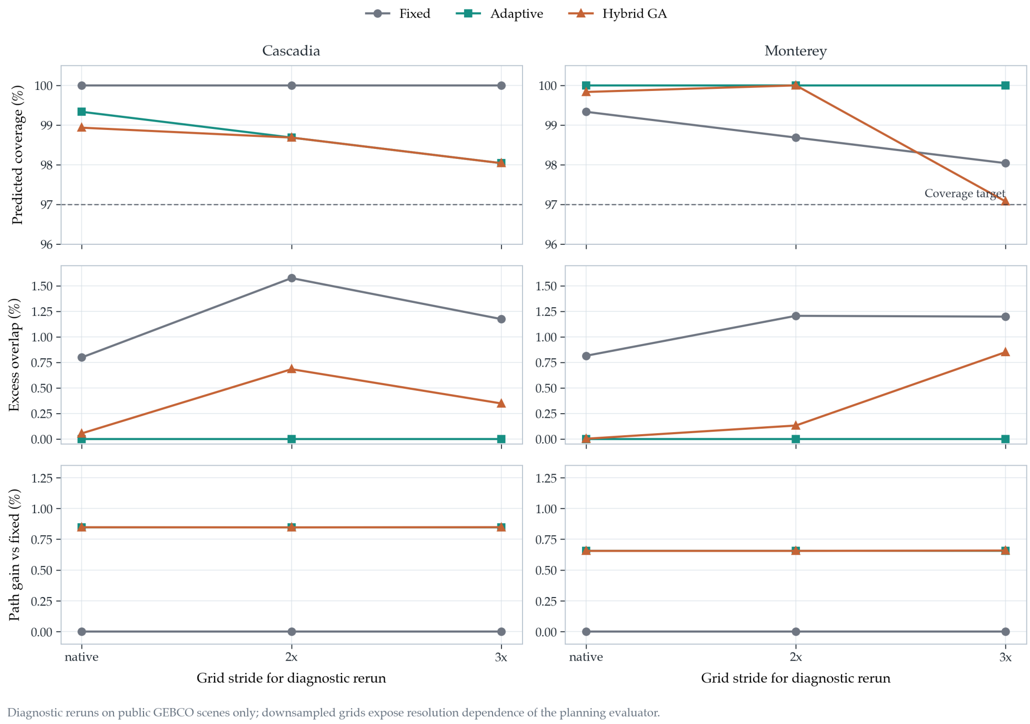

Figure 11. Public-scene grid-resolution sensitivity diagnostic. The same three public-scene methods are rerun on the native GEBCO-derived grids and on 2× and 3× strided downsampled grids using the reported evaluator. Hybrid GA reports seeds 0–4 for this diagnostic, whereas the main benchmark remains the native-grid 20-seed benchmark used in this study. The Monterey 3× case exposes a resolution-dependent coverage boundary for the Hybrid GA, so the diagnostic should be interpreted as a planning-evaluator sensitivity check rather than as field-performance evidence.

The supplementary prior-depth-bias and relief-scale checks test a simpler form of prior-map sensitivity. Rerunning the planners on a uniformly depth-biased prior (± 150 m) and on relief-scaled priors (0.7, 1.0, 1.3) left the public-scene layouts unchanged on Cascadia and Monterey: the dominant orientations and line-count modes stayed fixed, and the coverage, excess-overlap, and path-gain means were numerically unchanged. A separate objective-weight diagnostic reran the public scenes across a 3×3 coverage/overlap penalty grid and likewise retained the same selected line-family modes for the reported methods. The result does not make the weights universal, but it reduces the risk that the public result is a single penalty-weight artifact. This rules out trivial global prior offsets and local weight choices as the explanation for the public-scene result while motivating a separate execution-noise replay for spatially varying post-planning perturbations.

4.6. Execution-uncertainty Replay on Public Layouts

Figure 12 reports the Monte Carlo execution-uncertainty replay defined in the protocol. The replay uses the selected public-scene layouts and injects cross-track tracking error, heading error, and spatially correlated footprint-scale noise after planning. It therefore tests coverage and overlap margin under perturbed execution assumptions without re-optimizing the line family.

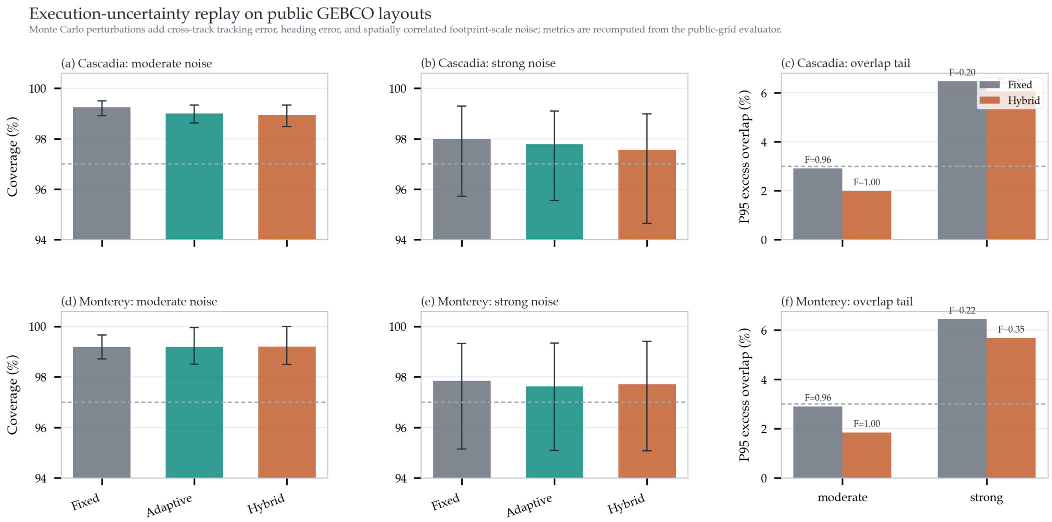

Figure 12. Execution-uncertainty replay on the two public GEBCO layouts. Panels (a), (b), (d), and (e) show mean predicted coverage with 5–95 percent Monte Carlo intervals under moderate and strong post-planning perturbations. Panels (c) and (f) show the 95th-percentile excess-overlap tail and the feasibility rate F for Fixed-Spacing and Hybrid GA. Under moderate perturbations, Hybrid GA remains feasible in 299/300 Cascadia trials and 300/300 Monterey trials; under strong perturbations, all methods lose substantial feasibility margin, indicating that execution-aware safety margins would be needed before field use.

The moderate-noise replay keeps the public layouts mostly feasible. On Cascadia, the Hybrid GA layout averages 98.94 percent predicted coverage with 1.99 percent 95th-percentile excess overlap and a 0.997 feasibility rate, compared with 99.25 percent coverage, 2.91 percent 95th-percentile excess overlap, and a 0.957 feasibility rate for Fixed-Spacing. On Monterey, Hybrid GA averages 99.20 percent coverage with 1.84 percent 95th-percentile excess overlap and a 1.000 feasibility rate, compared with 99.19 percent coverage, 2.90 percent 95th-percentile excess overlap, and a 0.963 feasibility rate for Fixed-Spacing. The strong-noise setting is deliberately harsher: feasibility falls to roughly 0.20–0.40 across methods and scenes. This result is useful because it both strengthens and limits the claim. The selected geometry-aware layouts retain useful margin under moderate execution perturbations, but stronger tracking and footprint errors would require explicit uncertainty-aware spacing margins rather than the nominal planner alone.

4.7. Independent USGS 30 m Public-grid Extension

To check whether the overlap-control pattern is only an artifact of the two regional GEBCO windows, we ran a separate extension on a higher-resolution public GeoTIFF from the USGS Southern Cascadia composite multibeam bathymetry release [2]. This extension is not mixed into the primary benchmark. It uses three automatically selected 4×5 nautical-mile crops from the USGS grid at low, medium, and high empirical terrain-complexity quantiles, and it repeats the same Fixed-Spacing, Adaptive Spacing without GA, and Hybrid GA comparison with 20 Hybrid GA seeds.

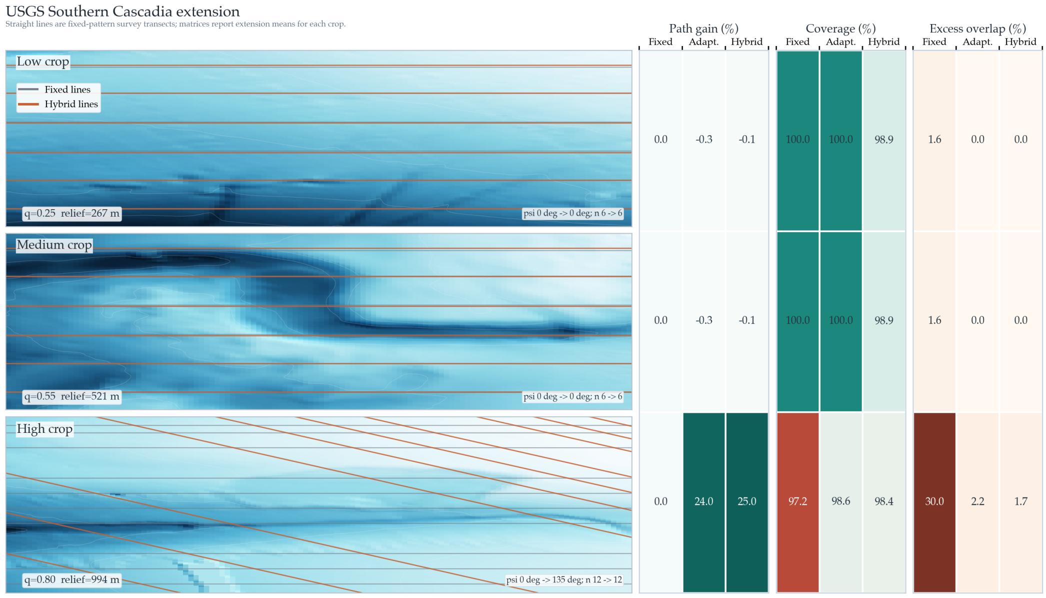

Figure 13. Independent USGS Southern Cascadia 30 m multibeam-derived public-grid extension. The left column shows the three selected crops with Fixed-Spacing and representative Hybrid GA line families, ordered by empirical terrain complexity from low to high. These plotted straight transects are the planned fixed-pattern survey-line families, not free-form vehicle trajectories. The three right-side matrices report benchmark means for path gain relative to Fixed-Spacing, predicted coverage, and excess-overlap violation recomputed from the extension CSV/JSON outputs. A wider map column preserves terrain readability while the matrices retain the scene-by-method comparison. The extension remains a public-grid numerical check, not an additional sea-trial or mission-log validation.

The extension supports a more specific interpretation of the contribution. On the low- and medium-complexity USGS crops, Fixed-Spacing was already feasible with 100.00 percent predicted coverage and 1.633 percent excess overlap, so terrain-aware planning mainly removed the small overlap surplus at a negligible path-length cost. On the high-complexity crop, however, Fixed-Spacing produced 29.960 percent excess overlap and was infeasible under the reported acceptance rule. Adaptive Spacing without GA shortened the path by 24.02 percent, reached 98.55 percent predicted coverage, and reduced excess overlap to 2.237 percent. Hybrid GA reached 98.44 percent predicted coverage, 1.732 percent excess overlap, and a 25.04 percent path gain. The extension therefore reinforces the same mechanism under a stricter public-grid setting: terrain-aware spacing matters most when the terrain genuinely stresses the fixed-spacing assumption, whereas easier public-grid crops should not be expected to show large route savings.

4.8. Coarse-prior to Fine-grid Public Replay

The preceding diagnostics still plan and evaluate on nominally matched grids, so we add a stricter public-grid replay on the same USGS source. Three USGS crops are selected by the same low/medium/high terrain-complexity policy, but the planner is given only coarsened prior surfaces with target cells of 120, 300, and 600 m. The resulting line family is then replayed on a finer 300×240 public grid with an effective cell size of about 31.0 m, without re-optimizing the heading or cross-track positions. This experiment is still a numerical public-data replay rather than a mission-log validation, but it tests whether the line layout survives a meaningful prior-resolution mismatch.

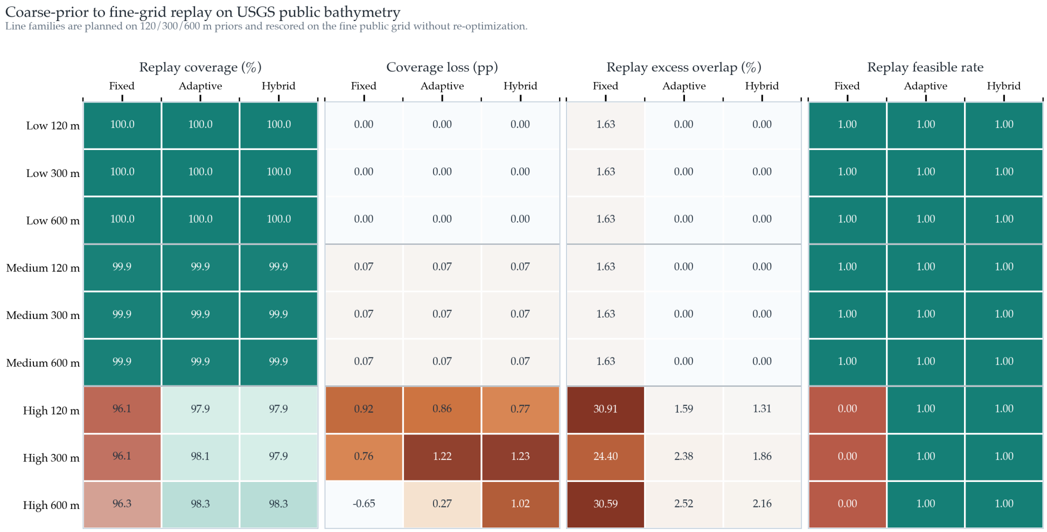

Figure 14. Coarse-prior to fine-grid replay on the USGS Southern Cascadia public bathymetry grid. Low, medium, and high-complexity crops are planned on 120, 300, and 600 m coarsened priors, then the same fixed-pattern line families are replayed on the fine public grid without re-optimization. The heatmaps report replay coverage, coverage loss relative to the coarse-prior prediction, replay excess-overlap violation, and replay feasibility. All entries are recomputed from `coarse_prior_replay/coarse_prior_replay_summary.csv`; Hybrid GA values are five-seed means for this diagnostic.

Figure 14 shows that the easy and medium USGS crops remain insensitive to this prior coarsening: all methods replay at 99.93–100.00 percent coverage, but Fixed-Spacing keeps about 1.63 percent excess overlap whereas the terrain-aware layouts remove that surplus. The high-complexity crop is more informative. Fixed-Spacing remains infeasible after replay at all three prior resolutions, with 96.06–96.32 percent coverage and 24.40–30.91 percent excess overlap. Adaptive Spacing without GA and Hybrid GA remain feasible on the fine replay grid at all three prior resolutions. Hybrid GA replays at 97.91–98.31 percent coverage, limits excess overlap to 1.31–2.16 percent, and gives 17.71–27.30 percent replay path gain relative to Fixed-Spacing. The result strengthens the mechanism claim without overextending it: terrain-aware spacing is stable under this controlled prior-resolution mismatch on the tested public grid, but the remaining 0.77–1.23 percentage-point Hybrid coverage loss on the high-complexity crop shows why survey-grade transfer still needs explicit uncertainty and prior-fidelity margins.

5. Discussion

Baseline limitation and what the GA actually adds.

The central question of this paper is whether explicit terrain-aware swath modeling improves offline fixed-pattern survey-line design when a prior bathymetric map is available. Across the two public GEBCO scenes, the most consistent advantage is better overlap control, together with a modest route shortening relative to Fixed-Spacing. At the same time, the public ablation shows that Adaptive Spacing without GA already captures nearly all of the public path-length gain. The Full Geometry-Aware Hybrid GA should therefore be framed mainly as a refinement layer that cleans up residual overlap and stabilizes the final layout, not as the sole source of public-scene efficiency.

Why the public gain is still meaningful.

The public result should not be judged by mean path shortening alone. The scene-level regime diagnostic shows why: both GEBCO scenes already begin below 1 percent fixed-baseline excess overlap, so there is limited room for dramatic route compression even though overlap cleanup remains useful. The stronger evidence is therefore the combined pattern across both scenes: Cascadia shows that terrain-aware spacing can regularize overlap without changing the sweep family, whereas Monterey shows that the same mechanism can trigger a genuine family rotation and a reduction from 73 to 59 lines while preserving high predicted coverage. For a fixed-pattern pre-mission planner, this combination is more informative than a single aggregate percentage because it reveals when the method is merely tightening an already reasonable baseline and when it is structurally changing the layout.

Public benchmark scope.

Using GEBCO subsets is a meaningful step beyond toy surfaces because the planner is tested on real public gridded seabed structure rather than only on hand-designed terrain. The two public scenes should be read as external-data evidence for the mechanism rather than as a claim of geographic representativeness across all continental margins, canyon systems, or survey products. GEBCO's own documentation treats the grid as a public information product derived from heterogeneous source data, and its disclaimer states that the grid should not be used for navigation or safety-at-sea decisions [1]. That statement is consistent with the role assigned here: the GEBCO layer is a reproducible planning benchmark input, while survey-grade mission products would be the natural transfer dataset for deployment studies. The planner assumes a prior bathymetric surface, nominally accurate execution, and idealized sensing geometry. The public GEBCO windows are also regional rather than sortie-scale, spanning roughly 128 $\times$ 178 km and 90 $\times$ 111 km after cropping, so the reported absolute route lengths should be interpreted as benchmark totals for large-area coverage rather than as single-sortie AUV mission budgets.

This scope also explains the role of the separate USGS public-grid extension. The extension tests whether the same overlap-regime interpretation survives a change in public data source, grid resolution, coordinate processing, and crop selection. Because the high-complexity USGS crop reproduces the high-overlap behavior seen in the synthetic stress case, while the easier USGS crops reproduce the small-gain behavior of the GEBCO scenes, the extension strengthens the mechanism argument without inflating the validation claim.

Sensitivity boundaries on declared sensor, map, and planning inputs.

The sensor-aperture, design-margin, and grid-resolution diagnostics sharpen the public-data interpretation rather than overturning it. Varying the MBES opening angle from 100° to 130° did not change the public-scene Hybrid GA heading or line-count modes in the five-seed diagnostic, which indicates that the main public layouts are not a fragile consequence of choosing exactly 120° in the reported evaluator. Varying the target-overlap margin from 10 percent to 20 percent also preserved feasibility on both public scenes, and the terrain-aware methods continued to suppress excess-overlap violation relative to fixed spacing. However, the selected orientations and line counts changed in both scenes, so the overlap margin should be treated as a declared survey-design parameter rather than as a universal constant. The grid-resolution diagnostic adds a second caution. Cascadia remains feasible under the tested downsampling levels, but Monterey becomes marginal for the Hybrid GA at 3$\times$ striding. This is consistent with Monterey's sharper canyon relief: reducing grid resolution changes the terrain gradients used by the swath evaluator and can erode coverage margin even when the nominal path length and line count remain similar. In practical pre-mission use, the prior bathymetry resolution should therefore be reported with the line plan, and coarse public grids should be treated as screening inputs rather than as a substitute for survey-grade terrain products.

The supplementary prior-depth-bias and relief-scale checks tighten this interpretation in a narrower way. Within the tested ± 150 m depth offsets and 0.7–1.3 relief scaling factors, the public-scene layouts and reported means do not move. That does not establish robustness to spatially structured map error, but it does show that the public-scene conclusion is not an artifact of small global prior shifts or uniform amplitude scaling.

The independent USGS 30 m extension and coarse-prior replay point in the same direction. They improve the evidence base because the data source, resolution, coordinate system, crop-selection policy, and prior-grid fidelity differ from the primary GEBCO benchmark. They also constrain the claim appropriately: two easier USGS crops show little route-length benefit, whereas the high-complexity crop shows a large improvement only because Fixed-Spacing is genuinely stressed by overlap violation. The coarse-prior replay adds a prior-fidelity stressor rather than a new field claim. On the high-complexity crop, Fixed-Spacing remains infeasible after replay on the fine grid, whereas Adaptive Spacing and Hybrid GA remain feasible at all three prior resolutions. The execution-uncertainty replay adds a different kind of check. Moderate cross-track, heading, and footprint-scale perturbations preserve nearly all public-layout feasibility, with Hybrid GA retaining a 0.997 feasibility rate on Cascadia and a 1.000 rate on Monterey. Stronger perturbations, however, reduce feasibility across all methods, so the replay supports the usefulness of geometry-aware layout margins while also showing why explicit uncertainty margins are needed before field execution.

Complex-terrain failure boundary.

The synthetic scenes are essential because they expose mechanism and boundary conditions that the public pair alone cannot. In the present dataset, the penalty of fixed spacing rises most sharply once the baseline line family enters a high-overlap regime, which in turn occurs on the more heterogeneous terrains. At the same time, the hardest complex-relief scene remains infeasible even after geometry-aware refinement. This negative result is scientifically important. It suggests that the present design space, which assumes a single global heading and a fixed parallel-line family, becomes too restrictive once relief varies strongly enough within the same region. In that regime, the limiting factor may no longer be line spacing alone, but the structural assumption that one globally oriented lawnmower family can represent the survey efficiently.

Deployment boundary and operation-relevant next steps.

Within this scope, the method is relevant to practice as a pre-mission design tool for settings where terrain knowledge already exists and the operator wants a simple fixed survey pattern for planning review, such as repeat mapping along shelf breaks, canyon margins, or corridor-style baseline surveys. In such cases, even modest route savings can matter if they are accompanied by better overlap control and predictable line configurations. At the same time, adjacent literature has moved toward richer localization, mapping, and field-oriented survey design: bathymetric SLAM studies increasingly couple exploration with loop-closure quality or robust bathymetric/range fusion [8,25], and field-oriented benthic planning studies use broad-scale bathymetric priors to generate informative initial AUV surveys under deployment constraints [9]. The present paper complements those directions by strengthening the fixed-line geometry that can be reviewed before sparse habitat sampling, map-consistency management, or closed-loop execution are introduced. The next transfer step is clear: replay the planner on additional survey-grade bathymetric products or mission logs, add uncertainty-aware spacing margins, and evaluate vehicle execution constraints such as roll/pitch/heave disturbance, current-driven track error, altitude-control error, tide or sound-speed uncertainty, and minimum-turn-radius limits.

Table 12 summarizes which interpretations are directly supported by the current benchmark and which limitations remain outside the manuscript's scope.

Table 12. Evidence-supported interpretations and remaining limitations of the numerical benchmark. Each row links one issue to the supporting evidence already reported in the manuscript and to the remaining boundary that should not be overclaimed.

Issue	Why it matters	Evidence from this study	Remaining boundary
Only two primary public scenes may not be enough.	A small public set cannot establish geographic representativeness.	The paper treats the two GEBCO scenes as the primary external-data benchmark, adds four supplemental GEBCO windows as a scene-selection check, uses synthetic terrains for mechanism and stress testing, and adds a separate USGS public-grid extension to test the same overlap-regime interpretation under a different public source.	No claim of representativeness across all seabed types or survey products.
Comparator scope is controlled.	Reviewers may ask why online SLAM, cooperative CPP, or energy-aware planners are not used as direct baselines.	The comparator ladder is confined to fixed-pattern line-layout variants evaluated by the same MBES terrain model, so the ablation isolates spacing, heading, and GA-refinement effects.	Broader autonomy, multi-vehicle, and dynamics-aware planners solve different coupled problems.
GEBCO is not survey-grade or navigation-grade.	Public global grids are heterogeneous information products, not raw MBES mission data.	The provenance table and GEBCO caveat state that the public layer is a reproducible prior-map benchmark.	Survey-log replay, sea-trial transfer, and safety-of-navigation use remain separate validation steps.
The public path-length gain is modest.	Route length alone does not capture the operational value of a survey-line layout.	The Results frame the public gain as overlap suppression plus stable fixed-line layout, with mean scene-level excess-overlap violation reduced from about 0.81 percent to 0.095 percent and Monterey showing a genuine family rotation with fewer lines.	Large route savings are expected mainly when the fixed-spacing baseline enters a high-overlap regime.
GA adds little beyond adaptive spacing.	The ablation shows that adaptive spacing captures nearly all public route shortening.	The contribution is reframed: GA is a refinement layer for residual-overlap cleanup and seed-level stabilization.	No claim that GA alone is the dominant innovation or globally optimal.
The complex-terrain case fails.	A hidden infeasible case would undermine robustness claims.	The failure is retained as a boundary-of-validity result linked to the single-heading lawnmower assumption.	Strongly heterogeneous terrain may require segmented headings, local replanning, or richer route topology.
Sensitivity remains expandable.	Planning depends on sensor aperture, overlap margin, grid resolution, prior-map fidelity, and execution error.	Public diagnostics test MBES opening angle, target overlap, simple prior-map perturbations, grid-resolution changes, coarse-prior/fine-grid replay, execution-noise replay, and an independent USGS 30 m public-grid extension.	Mission-log replay, vehicle dynamics, and sonar calibration are the priority validation steps before operational deployment claims.

6. Conclusion

This paper studied terrain-aware pre-mission survey-line planning for AUV multibeam mapping under a fixed lawnmower traversal when a prior bathymetric model is available. The main challenge is that effective MBES swath width varies with local seafloor geometry, so uniform line spacing creates avoidable overlap in some regions while reducing coverage margin in others. To address this issue, the paper combined a terrain-aware sensor–terrain model with a two-stage planner consisting of a global orientation search and a local line-placement refinement stage.

On the two public GEBCO scenes in the benchmark reported here, the Full Geometry-Aware Hybrid GA reached a predicted coverage range of 98.97–99.63 percent, reduced mean scene-level excess-overlap violation from about 0.81 percent to 0.095 percent, and achieved a 0.75 percent relative shortening of mean scene-level path length compared with Fixed-Spacing. The strongest public effect is therefore not dramatic path compression but better overlap discipline under the same fixed traversal. Cascadia behaves mainly as an overlap-cleanup control case, whereas Monterey shows a structural rotation from 0° to 90° together with a reduction from 73 to 59 lines. The scene-level regime diagnostic further shows why the public route-length gains are modest: both GEBCO scenes already begin below 1 percent fixed-baseline excess overlap, so they do not occupy the near-failure regime in which the largest geometry-aware savings appear. The public-scene ablation further shows that Adaptive Spacing without GA already captures nearly all of the route shortening, so the hybrid GA is best interpreted as a stable refinement mechanism rather than as the sole driver of efficiency. The 20-seed public-scene runs converged to identical heading and line-count modes on both scenes, indicating strong seed-level repeatability under the present configuration.

The remaining diagnostics define the transfer conditions. MBES opening-angle, target-overlap, and simplified prior-map perturbation checks retain the main native-grid interpretation under the tested public-scene settings. The grid-resolution diagnostic adds an explicit resolution boundary, including a marginal Monterey Hybrid GA result under $3\times$ grid coarsening. The coarse-prior/fine-grid replay adds a stronger public-grid prior-fidelity test: on the USGS high-complexity crop, terrain-aware layouts remain feasible when planned from 120–600 m priors and replayed on the fine grid, whereas Fixed-Spacing remains infeasible. The execution-uncertainty replay shows that the selected public layouts retain high feasibility under moderate cross-track, heading, and footprint-scale perturbations, while stronger perturbations erode feasibility across all methods. The separate USGS 30 m public-grid extension strengthens the mechanism evidence on a higher-complexity crop while confirming that easier public crops may show little route-length benefit. Across the combined evidence, the largest gains occur when the Fixed-Spacing baseline enters a high-overlap regime, but the hardest complex-relief scene still remains below the 97 percent coverage target under the present single-heading lawnmower formulation.

Taken together, these results support a specific claim: geometry-aware spacing improves pre-mission fixed-pattern AUV survey-line planning on public bathymetry benchmarks, and GA refinement makes the resulting layout more repeatable and better controlled with respect to residual overlap. The present contribution is a reproducible public-bathymetry benchmark study with an explicit data-provenance trail, a measured role for GA refinement, a documented complex-terrain failure case, a higher-resolution multibeam-derived public-grid extension, a coarse-prior/fine-grid replay, and a first execution-noise replay. Natural next steps are to test the planner on additional survey-grade datasets or mission logs, introduce uncertainty-aware and execution-aware constraints, broaden the baseline set, and examine whether the same ranking persists when moving from public bathymetry benchmarks toward operation-relevant AUV survey planning.

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Abbreviations

The following abbreviations are used in this manuscript:

AUV	Autonomous Underwater Vehicle
CPP	Coverage Path Planning
GA	Genetic Algorithm
GEBCO	General Bathymetric Chart of the Oceans
MBES	Multibeam Echo Sounder
USGS	United States Geological Survey

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